

Tandem tapered roller bearings no-load torque loss in a rear axle gear transmission

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ARTICLE INFO

Keywords:

Friction
Measurement
Roller bearings
Lubrication oil

ABSTRACT

The main no-load power loss source of a rear axle is the pre-loaded pinion tapered roller bearings, as concluded in a previous work. Therefore, the pinion roller bearings torque loss was studied in detail, where the influence of several parameters was considered, such as, the starting torque of the pinion tapered roller bearings, the oil level, the lubricant viscosity and the rotational speed.

Due to the nature of the assembly of the pinion shaft assembly, pre-load in the tapered roller bearings is difficult to measure, however the starting torque of this assembly can be easily measured.

For this reason, in this work a rolling bearing torque loss empirical equation was developed, where it uses as input parameters the starting torque, the oil dynamic viscosity, operating speed and the rolling bearing mean diameter and width.

1. Introduction

Energy efficiency in vehicles is becoming an increasingly important aspect, in order to reduce gas emissions and operating cost. The vehicle drivetrain is responsible for a significant part of the overall power loss, where one of its most relevant components, the axle box, features a low efficiency, between 90 and 93% [1].

Therefore the power loss of a rear axle gear transmission has become an important field of investigation [2–8]. Some studies with tapered roller bearings lubricated with axle oils have been done [9,10], aiming to analyse the influence of lubricant and operating conditions in the power loss of such rolling bearings.

A previous work [11] allowed to conclude, that the pinion tapered roller bearings are the main responsible for the rear axle no-load inefficiency, because of the high preload that they are subjected to, necessary to assure a high rigidity to the pinion gear set, as requested by the hypoid gear set.

The power loss in rolling bearings can be divided in two major parts, the load dependent and load independent losses.

The load dependent losses, which are outside of the focus of this work, occur due to the friction between contacting bodies of power transmitting components. For the rolling bearings, the load dependent losses depend on their type and size, on rolling and sliding conditions

and on lubricant type [12].

The no-load losses occur due to the mechanical components immersed, partially or totally, in the oil sump, even without torque transmission. The torque loss generated by tapered roller bearings also depends on their size, on the sliding and rolling conditions and on the lubricant type.

Studies to the rolling bearings losses have already been done, aiming to develop models that can predict the torque losses, usually based on experimental results. The SKF [13] model is complex and considers the influence of a lot of parameters, being one of the most used models due to its reasonably good estimations and high versatility, although it is the more complex of the models. The ISO [14] model is simpler and depends on less parameters, mainly on the bearing mean diameter, lubricant viscosity, rotational speed, equivalent load and some coefficients, where its simplicity leads to less satisfactory results. In 1972 Witte [15] proposed a model for the estimation of the torque loss of axially loaded tapered roller bearings, a model that was specially design for the same type of bearings as in the current study. A theoretical equation for the friction torque loss was derived for this specific case of study, necessary to fundament some of the presented work. Since the current work also deals with axially loaded tapered rolling bearings Witte's model may be the model with more potential. All the models depend on the preload that the rolling bearings are subjected to, a variable difficult to measure.

Therefore, this work will be focused on torque loss of pre-loaded

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<https://doi.org/10.1016/j.triboint.2021.106876>

Received 11 November 2020; Received in revised form 30 December 2020; Accepted 9 January 2021

Available online 15 January 2021

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Nomenclature

α, β, θ	equation exponents (–)
η	dynamic viscosity (N s m^{-2})
μ	lubricant viscosity (cp)
μ_{bl}	boundary film coefficient of friction (–)
ω	angular velocity (rad s^{-1})
ω_0	reference angular speed (rad s^{-1})
ρ_{oil}	oil density (kg m^{-3})
ν	kinematic viscosity ($\text{mm}^2 \text{s}^{-1}$)
a_{lub}	lubricant level from the shaft center (m)
C_m	dimensionless drag torque (–)
C_m^f	dimensionless constant (–)
$C_{1,2}$	equation constants (–)
d_m	rolling bearing mean diameter (mm)
d_{sh}	shaft diameter (m)
F_a	axial load (N)
F_a^*	axial load (lb)
Fr	Froude number (–)
G	bearing geometric factor (–)
g	gravitational acceleration (m s^{-2})
h	oil immersion depth (m)
M_0	no-load dependent torque loss (Nmm)
M_1	load dependent torque loss (Nmm)
M_{drag}	frictional moment of drag losses (Nmm)
$M_{hydrodynamic}$	rolling bearing hydrodynamic torque loss (Nm)
M_{rr}	rolling frictional moment (Nmm)
M_{seal}	frictional moment of the seals (Nmm)
M_{si}	sliding frictional moment (Nmm)
M_{st}	rolling bearing starting torque (Nm)
$M_{transient}$	rolling bearing transient torque loss (Nm)

M_{Witte}	torque loss according to Witte (lb-in)
n	rotational speed (rpm)
R_m	rolling bearing mean radius (m)
r_o	gear outer radius (m)
Re_f	Reynolds number - Fossier model (–)
Re_r	Reynolds number - Rolling bearing friction torque loss model (–)
S_1, S_2	sliding friction moment (–)
S_m	immersed surface area of the pinion (m^2)
T	lubricant temperature (degrees)
$T_{1,2}$	rolling bearing width (m)
$T_{V0}^{P,asb}$	pinion assembly torque loss (Nm)
$T_{VD,Simrit}$	seal torque loss (Nm)
$T_{VD,Simrit}^{P,sl}$	pinion seal torque loss according to the Simrit model (Nm)
$T_{VD}^{P,sl}$	pinion seal torque loss (Nm)
$T_{VL0,Exp}^{P,rb}$	experimental pinion tapered roller bearings torque loss (Nm)
$T_{VL0,ISO}$	ISO rolling bearings friction moment (Nmm)
$T_{VL0,SKF}$	SKF rolling bearings friction moment (Nmm)
$T_{VL0,Witte}$	Witte rolling bearings friction moment (Nm)
T_{VL0}	no-load rolling bearing torque loss (Nm)
$T_{VL0}^{P,rb}$	pinion tapered roller bearings torque loss (Nm)
$T_{VL0,FS}$	Fossier churning torque loss (Nm)
$T_{VZ0,FS}^P$	pinion gear churning torque loss according to the Fossier model (Nm)
T_{VZ0}^P	pinion gear churning torque loss (Nm)
V_0	oil volume (m^3)
Y	axial load factor (–)

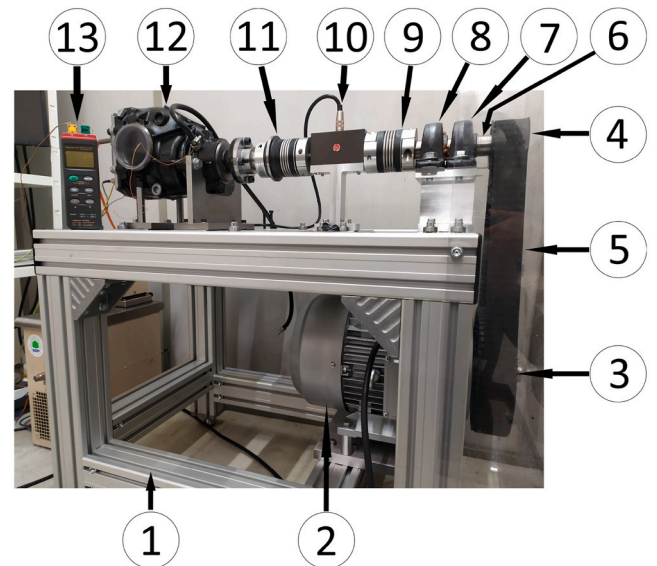
tapered roller bearings, where experimental tests will be performed and the different influences in the torque loss will be analysed, such as the pre-load, rotational speed, lubricant viscosity and level. The experimental results will be compared to the models estimations, and then, an empirical equation for the friction torque of pre-loaded tapered roller bearings in tandem will be developed based on the measurements of the starting torque. Unlike the axial force due to the pre-load, the starting torque can be directly measured. The lubricant level and viscosity, rotational speed and rolling bearings geometrical specifications will be also considered as variables of the equation.

To conclude, the sensitivity of the developed equation to parameters such as, speed, load, lubricant level and viscosity is analysed and compared with the SKF [13] and Witte [15] models. This is done in order to assure that there is an agreement between the developed and a current state of the art rolling bearing torque loss models.

2. Test rig and operating conditions

2.1. Test rig

The test rig, shown in Fig. 1, is composed of a supporting structure (1) made of aluminium profiles, an AC motor (2) with speed control by frequency variation, a timing pulley (3) mounted on the motor output shaft and a second timing pulley (4) mounted on the upper shaft and a timing belt (5). The upper shaft (6) is supported by two ball bearings (7, 8) and it is connected to the test axle box (12) by two spring couplings (9, 11) and a torque transducer (10). The spring coupling (11) is a limiting torque coupling. The upper axle line, including the shaft (6), the rolling bearings (7, 8), the torque transducer (10) and the axle box (12) have been aligned using high precision laser technology, in order to



Key

1- Structure;	6- Shaft;	11- Limiting torque coupling;
2- AC motor;	7,8- Rolling bearings;	12- Rear axle;
3,4- Timing puley;	9- Spring coupling;	13- Thermologger;
5- Timing belt;	10- Torque transducer;	

Fig. 1. Test rig.

minimise any misalignment that could interfere with the torque measurements.

The axle box (or differential, 12) is from a rear wheel drive vehicle with reference 33107518807. It is an open differential with a reduction ratio $i = 1/3.07$. The specifications of the crown and pinion gears, of the sun and planet, of the rolling bearings and of the seals used in this axle are given in Tables 1–3.

A torque transducer (10) from ETH (reference DRDL-II-50-10-n), continuously measures the torque and speed transmitted to the axle box. The torque transducer specifications are presented in Table 4.

A thermologger (13) measures the temperatures of the axle box oil bath and of the oil trapped in the pinion shaft rolling bearings chamber, shown in Fig. 2. The thermologger uses two type K thermocouples to measure the temperature. The thermologger specifications are presented in Table 5.

The axle box (12) is lubricated with a SAE 75W90 commercial axle gear oil, Table 6. The oil level volume recommended by the axle box manufacturer was assured, which represents ≈ 1 L with the crown wheel mounted and ≈ 1.5 L without the crown wheel, corresponding to a 20 mm distance from the pinion shaft axis and the oil level.

2.2. Operating conditions

To understand the influence of different operating conditions on the tapered roller bearings (no-load) torque loss, the tests were performed for large ranges of operating speed, of the oil sump temperature, of the preload applied to tapered roller bearings of the pinion shaft assembly and of oil sump level.

2.2.1. Speed and temperature ranges

The operating rotational speeds selected were 500, 750, 1000, 1500, 2000, 2250 and 2500 rpm. These speeds adequately replicate the real operating speeds of the axle box, corresponding to vehicle linear speeds from 20 to 100 km/h moving forward. Additional experimental tests were performed, so a more detailed analysis could be done for tapered roller bearing behaviour at a very low speeds (86, 130, 200, 300, 400, 500 and 750 rpm).

The oil sump temperatures in the Gear chamber (T_{sump}^{Gear} , see Fig. 4) were selected in the range between 50° and 90 °C, assuring identical oil viscosity intervals, as shown in Fig. 3. The tests were performed at the following viscosities and temperatures: 66.8 cSt @ 50.0 °C, 58.9 cSt @ 53.3 °C, 50.9 cSt @ 57.3 °C, 42.9 cSt @ 62.2 °C, 34.9 cSt @ 68.5 °C, 27.0 cSt @ 77.1 °C, 19.1 cSt @ 90 °C.

2.2.2. Starting torque and preload

The preload is applied by tightening the nut that presses the two tapered roller bearings one against the other, as shown in Fig. 4, up to a predefined torsion torque. However, it is difficult to measure directly the preload, as it would be necessary to install strain gauges on the pinion shaft to measure the axial elastic deformation of the shaft and calculate from it the axial preload.

Nevertheless, it is possible to measure the starting torque generated by the tapered roller bearings by the application of the preload. The starting torque is the torque needed to initiate the rotation of the shaft

Table 1
Geometrical parameters of the differential gears.

Parameter	Pinion	Crown Wheel	Sun Gear	Planet Gear
Number of Teeth	14	43	17	10
Gear ratio	1/3.07	-	-	-
Outer Diameter/mm	93	184	79	54
Inner Diameter/mm	65	130	51	36
Pitch Diameter/mm	79	157	-	-
Module/mm	4	4	-	-
Width/mm	39	15	-	-

Table 2
Rolling bearings in the differential.

Location:	Pinion	Input coupling	Output shaft
Actual Rolling Bearing	Koyo HM 801349/ 310	Koyo HM 88542/ 510	FAG F-801298- TR1P -H79-T29
Quantity	1	1	2
Bore Diameter/mm	40.48	31.75	47.625
Outside Diameter/mm	82.55	73.025	88.9
Width/mm	29.37	29.37	20.638
Number of rollers Z/-	19	18	19
Mean diameter dm/mm	61.515	52.39	68.263
Roller mean diameter D/mm	7.72	6.67	9.98
Roller length l/mm	22.2	21.5	14.5
Contact angle α° /degree	20	20	12
Geometry coefficient γ /-	0.118	0.120	0.14
Geometry coefficient γ^* /-	1.74	1.74	1.68
Half width of the RE contact b/mm	≈ 3.87	≈ 3.52	≈ 4.25
Roller - rib contact width f/mm	≈ 1.2	≈ 1.1	≈ 1.2
Half included roller angle ν /degree	4.93	4.93	3.13
Geometry coefficient B/-	0.163	0.162	0.112
Geometry coefficient K' /-	0.0569	0.0578	0.0354
Geometry coefficient K/-	0.0612	0.0621	0.0396

Table 3
Seals used in the differential.

Location:	Input coupling	Output shaft
Actual Seal	Corteco 01027755B	Corteco 01033294B
Quantity	1	2
Shaft Diameter/mm	45	44
Housing Diameter/mm	75	90.1

Table 4
Torque transducer specifications.

Torque Transducer Type ETH DRDL-II-50-10-n	
Specification	Value
Nominal torque/Nm	50
Measurement range/Nm	10; 50
Non-linearity/%	< 0.1
Hysteresis/%	< 0.1
Accuracy/%	0.01
Temperature sensitivity/% K	0.01

(onset of movement), which can be measured accurately using a torque wrench (see Fig. 5). Furthermore, the starting torques increases as the nut is increasingly tightened and the preload increases, although such variation is not linear.

In this case, the starting torque was measured using a TOHNICHI DB6N torque wrench, which has a measuring range between 0.6 Nm and +6 Nm and an accuracy of $\pm 3\%$, Fig. 5.

2.2.3. Rolling bearings immersion depth

The lubricant level have a major effect on the rolling bearing drag torque loss, so its influence will be analysed by changing to the oil sump level, as shown in Fig. 4. Four oil levels of $a_{lub} = 20, 21, 22$ and 24 mm were selected, allowing its influence in the torque loss to be analysed, without compromising the rolling bearings lubrication, where the variable a_{lub} represents the distance between the oil surface and the shaft centre.

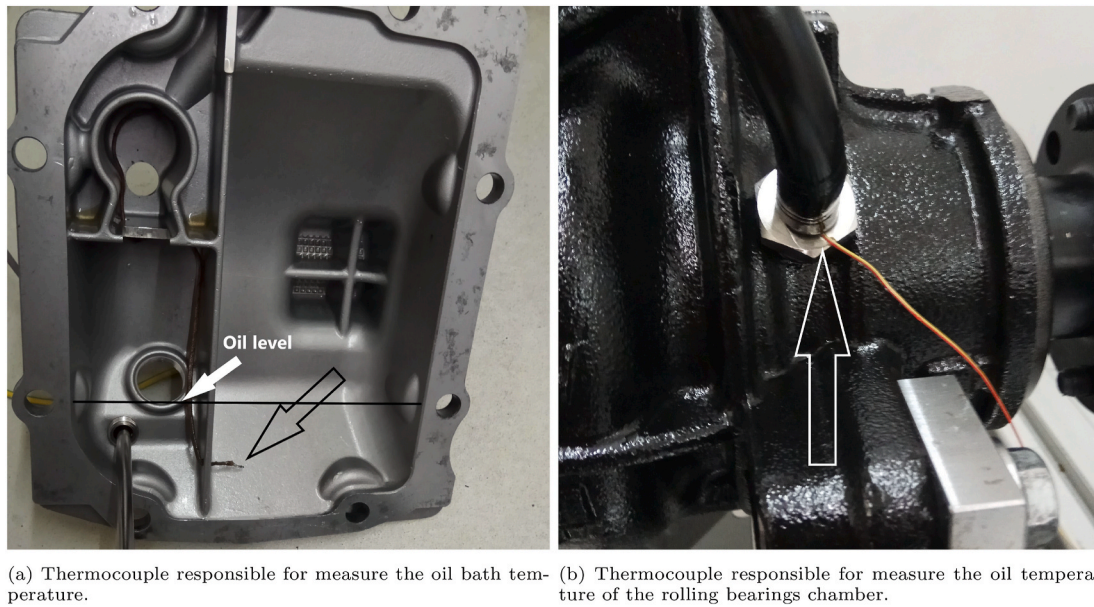


Fig. 2. Temperature measure system.

Table 5
Thermologger specifications.

OMEGAETTE HH306 Thermologger		
Specification	Range	Value
Measurement range	-	-200 °C ~ 1370 °C
Resolution	-200 °C ~ 200 °C	0.1 °C
	200 °C ~ 1370 °C	1 °C
Accuracy	-200 °C ~ 200 °C	±(0.2% reading + 1 °C)
	200 °C ~ 400 °C	±(0.5% reading + 1 °C)
	400 °C ~ 1370 °C	±(0.2% reading + 1 °C)
Sample rate	-	1.25 Hz

Table 6
SAE 75W-90 commercial axle gear oil characteristics.

Test	Unit	Test method	Result
Volumetric mass at 15 °C	kg m ⁻³	DIN 51757 D	866
Viscosity at 40 °C	mm ² s ⁻¹	ASTM D445	101
Viscosity at 100 °C	mm ² s ⁻¹	ASTM D445	15
Viscosity index	-	ASTM D2270	157
Viscosity Brookfield at -40 °C	mPa s ⁻¹		66 000
Flash point	°C	ASTM D97	-51
Pour point	°C	ASTM D92	190

The tapered roller bearings (REB) chamber is connected to the pinion/crown wheel (gear) chamber by two large channels, allowing the oil to circulate freely between the two chambers. Furthermore, the oil in the REB chamber can pass through the each bearing and return to the oil sump, assuring a proper lubrication of the tapered roller bearings.

The oil sump level is not perturbed by the pinion gear rotation, because the pinion immersion depth is small, the radius of the pinion is also small and the pinion rotation speed is limited to 2500 rpm.

2.3. Experimental methodology

The aim of this work is to measure the pinion assembly torque loss, the crown assembly was removed from the axle box.

During the no-load torque loss tests of the pinion assembly, all the operating conditions were permanently measured and recorded. The

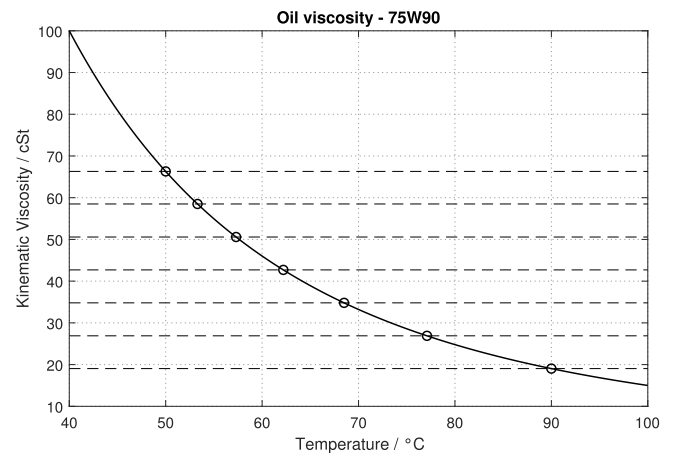


Fig. 3. Selected temperatures over the kinematic viscosity variation.

axial pre-load that defines the starting torque, is imposed through the nut located in the tapered roller bearings shaft, as shown in Fig. 4. The input speed is set by the frequency controller of the AC motor, with an accuracy of ± 1.0 rpm.

The oil sump temperature in the Gear chamber (T_{sump}^{Gear}) is selected (between 50 °C and 90 °C), and the electric resistance immersed in the oil sump of the Gear chamber heats up the oil up to that temperature and keeps it constant (± 1.0 °C). There is no cooling capability. The test starts when the oil sump temperature in the Gear chamber (T_{sump}^{Gear}) reaches the selected value. The temperature in the REB chamber is free.

At low speeds ($n \leq 1500$ rpm) and at high oil sump temperatures ($T_{sump}^{Gear} \geq 77.5$ °C) it is possible to keep the temperature in the Gear chamber constant, whatever the test speed. In this situation, the temperature in the REB chamber is similar to the temperature in the Gear chamber. The torque loss and the temperatures in both chambers are recorded when those temperatures reach a stabilized value, that is, a temperature variation of less than ± 1.0 °C during 5 min.

At high speeds ($n \geq 2000$ rpm) and low oil sump temperatures ($T_{sump}^{Gear} \leq 68.5$ °C) it is not possible to keep the temperature in the Gear chamber constant. In this case, the temperature in the REB chamber is higher than the temperature in the Gear chamber. The torque loss and the

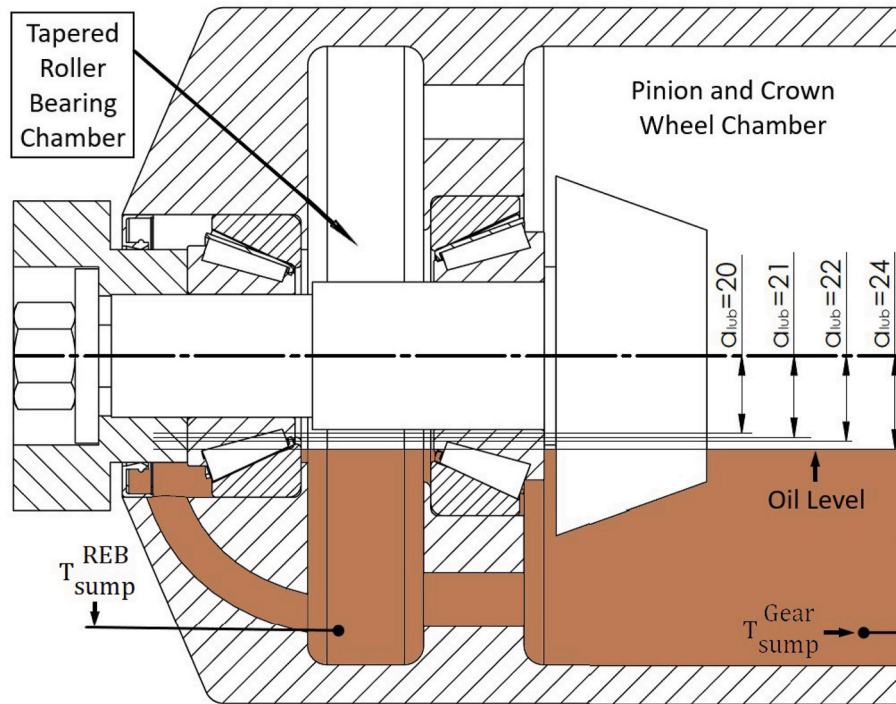


Fig. 4. Schematic view of the tapered roller bearings chamber with oil sump lubrication.



Fig. 5. Torque wrench used to measure the starting torque [16].

temperatures in both chambers are recorded when those temperatures reach a stabilized value, that is, a temperature variation of less than $\pm 1.0^\circ\text{C}$ during 5 min.

Due to the differences in size and oil volume, and to the power loss in the tandem tapered roller bearings, it is not possible to keep the temperature constant in both test chambers. So, it was decided to control and keep constant the temperatures in the Gear chamber, when the

operating conditions allow that. Later on, those oil sump temperatures in the Gear chamber will be used for the tests with the complete axle box and with the crown gear alone.

The tests were performed under dip lubrication, and the oil level was set between $a_{lub} = 20$ mm up to 24 mm, as shown in Fig. 4. This way it was possible to measure the pinion assembly torque loss ($T_{V0}^{P,asb}$, Fig. 6), that includes the pinion gear churning ($T_{VZ0}^{P,g}$), tapered roller bearings ($T_{VL0}^{P,rb}$) and seal torque loss ($T_{VD}^{P,sl}$), as expressed in equation (1).

$$T_{V0}^{P,asb} = T_{VZ0}^{P,g} + T_{VL0}^{P,rb} + T_{VD}^{P,sl} \quad (1)$$

3. Experimental results

3.1. Starting torque measurements

The starting torque was measured using the torque wrench shown in Fig. 5. In order to properly measure the starting torque promoted by the pinion assembly tapered roller bearings, the crown assembly and the oil sump were removed (the bearings have been previously oiled) and the

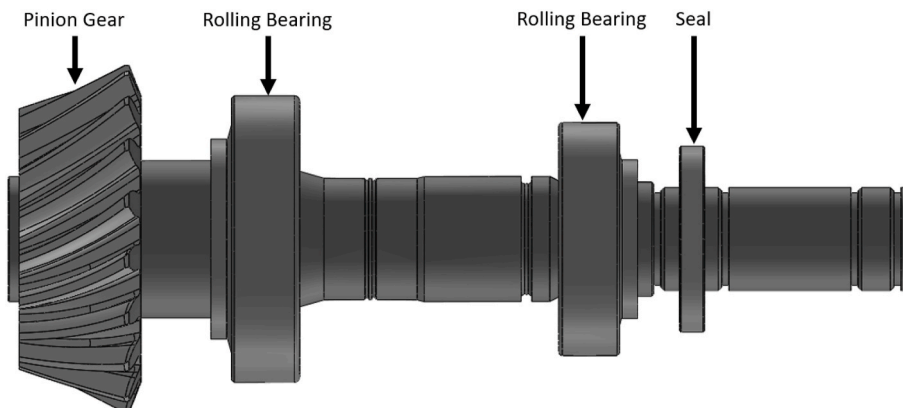


Fig. 6. Schematic view of the pinion assembly; tapered roller bearings (REB), pinion gear and seal.

differential was at rest (still). The measurements were done the ambient temperature ($\approx 20^\circ\text{C}$). Three different starting torques had been applied by tightening the shaft nut, respectively, 0.75 Nm, 1.90 Nm and 3.00 Nm. These starting torques represent the sum of the starting torques generated by each one of the two tapered roller bearings, assembled in a back-to-back arrangement and the pinion seal.

3.2. Measurements at low speeds up to 750 rpm

3.2.1. Temperature measurements at low speed up to 750 rpm

Fig. 7 shows the stabilized temperatures in the REB and Gear chambers, for operating speeds up to 750 rpm. Under these conditions, the oil sump temperature in the Gear chamber generally remains almost constant ($\pm 1.0^\circ\text{C}$) independently of the operating speed and of the starting torque.

At these speeds, the oil sump temperature in the REB chamber is always lower than the temperature in the Gear chamber ($T_{\text{sump}}^{\text{REB}} \leq T_{\text{sump}}^{\text{Gear}}$), but it increases progressively as the speed increases. However, for this speed range, $T_{\text{sump}}^{\text{REB}}$ is almost independent of the starting torque.

These measurements show that it is not possible to keep the oil sump temperature constant in both chambers simultaneously.

3.2.2. Torque loss measurements at low speed up to 750 rpm

In order to better understand the influence of the starting torque, M_{st} , on the pinion assembly, it was necessary to measure and analyse the torque loss measurements at low speeds ($0 \leq n \leq 750$ rpm), as shown in Fig. 8.

At very low speed ($\omega < 3\pi$ rad/s) the torque loss decreases sharply as the speed increases, corresponding to what can be designated as a transient torque, corresponding to the transition from starting to dynamic conditions.

Simultaneously, as the speed increases ($\omega > 3\pi$ rad/s, minimum speed allowed by the test machine) the torque loss increases monotonically with the speed, corresponding to a hydrodynamic behaviour.

The overall torque loss of the pinion assembly is the sum of these two components, transient and hydrodynamic, as shown schematically in Fig. 9. However, Fig. 8 also shows that the hydrodynamic torque depends on the starting torque, since higher starting torques always generate higher hydrodynamic torques, even in the operating speed range ($500 \leq n \leq 2500$ rpm) of the axle box.

At very low speeds, it is possible to observe a clear decrease of the transient torque, up to a point the hydrodynamic torque becomes similar to the transient torque.

3.3. Measurements in the range 500–2500 rpm

3.3.1. Temperature measurements in the range 500–2500 rpm

Fig. 10 shows the stabilized temperatures in the REB and Gear chambers, for operating speeds in the range 500–2500 rpm.

The oil sump temperature in the REB chamber, $T_{\text{sump}}^{\text{REB}}$, increases as the operating speed and/or the starting torque increase. In fact, the power loss in the tandem tapered roller bearings heats up the oil sump in the REB chamber. Since the oil can flow between the two chambers, the hotter oil coming from the REB chamber also heats up the colder oil in the Gear chamber. However, the oil volume in the REB chamber is significantly smaller than the oil volume in the Gear chamber and, consequently, the temperature increase in the Gear chamber is significantly lower than in the REB chamber.

At high temperatures ($68.5^\circ\text{C} \leq T_{\text{sump}}^{\text{Gear}} \leq 90.0^\circ\text{C}$), the oil sump temperature in the Gear chamber remains almost constant ($\pm 1.0^\circ\text{C}$), independently of the operating speed and of the starting torque. In this temperature range, the hot oil flowing from the REB chamber does not

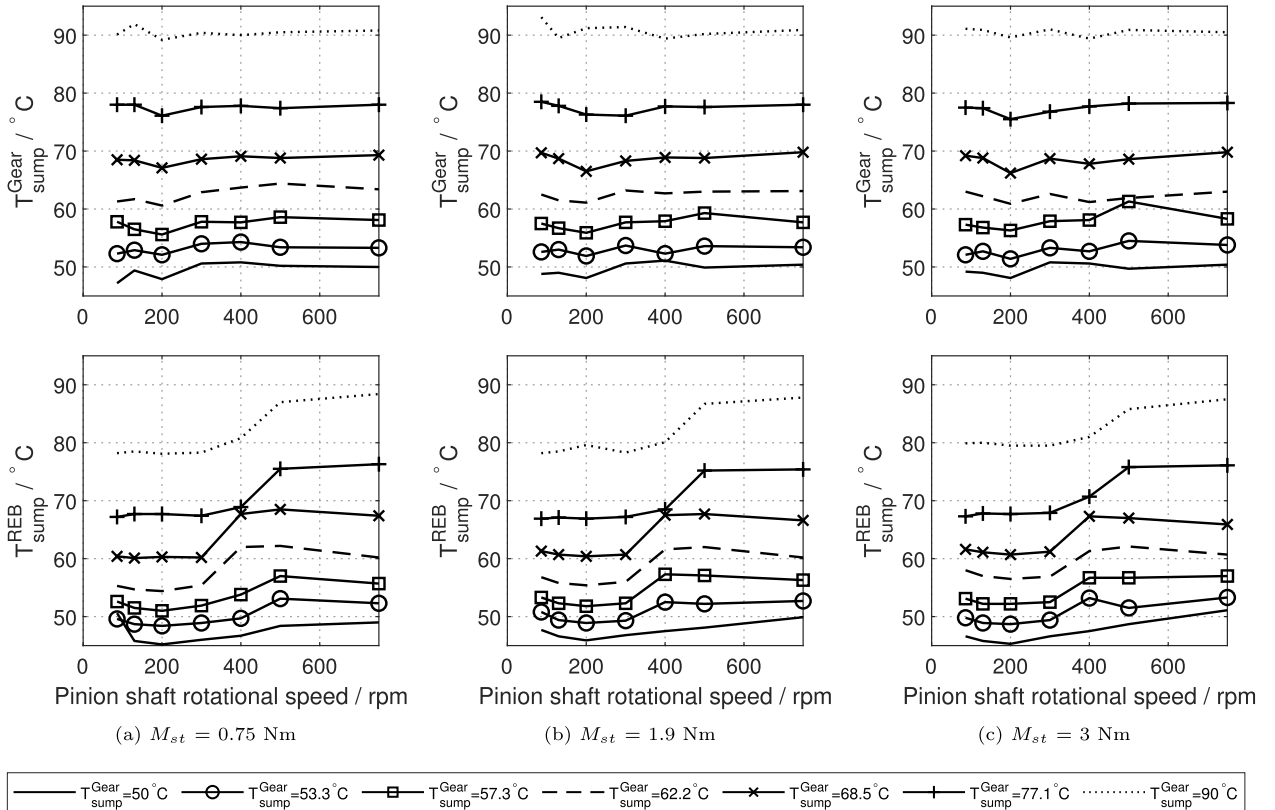


Fig. 7. Stabilized temperatures in the REB and Gear test chambers for different starting torques at low speeds.

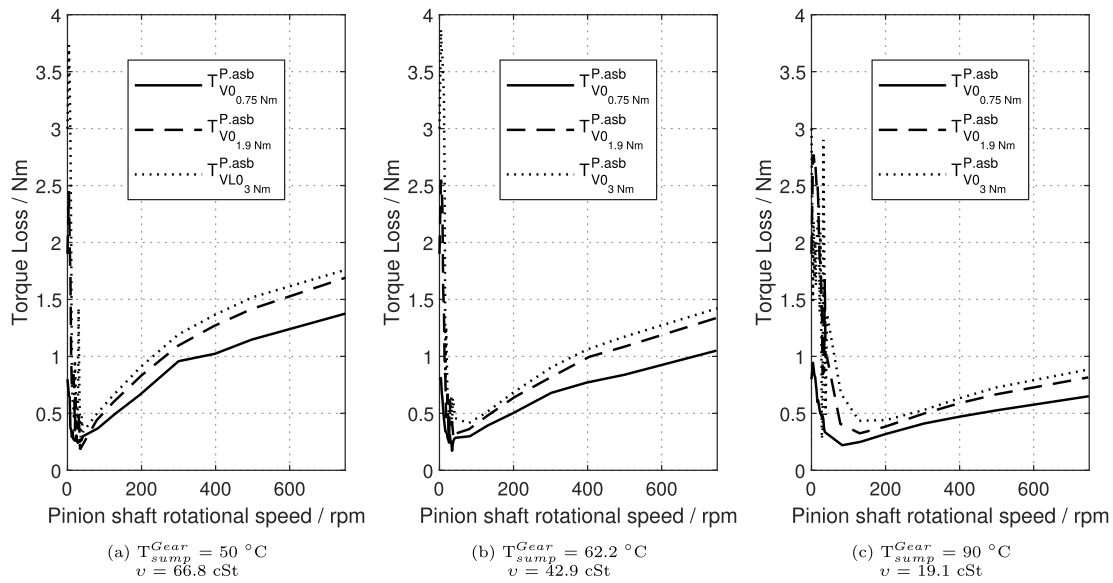


Fig. 8. Pinion assembly torque loss for different starting torques at low speeds.

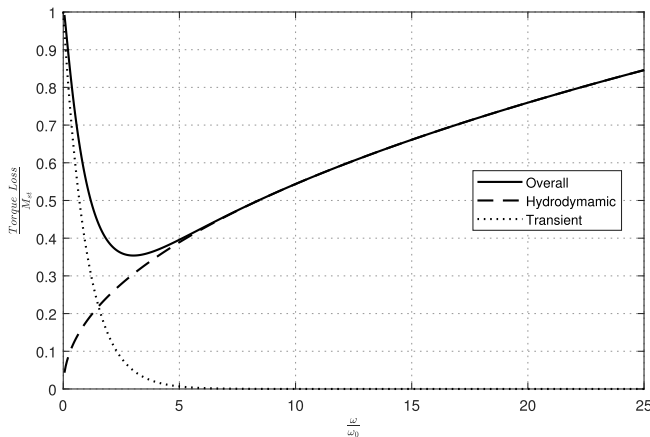


Fig. 9. Schematic representation of the transient and hydrodynamic torques.

affect the oil sump temperature in the Gear chamber. At lower temperatures ($50.0\text{ }^{\circ}\text{C} \leq T_{\text{Gear_sump}}^{\text{Gear}} \leq 62.2\text{ }^{\circ}\text{C}$), the oil sump temperature in the Gear chamber heats up as the speed and/or the starting torque increase, although the influence of the starting torque is small ($+2/4\text{ }^{\circ}\text{C}$ at 2500 rpm).

As an example, Fig. 10 indicates that, at 1500 rpm and 1.9 Nm, the oil sump temperature in the Gear chamber remains constant at $57.3\text{ }^{\circ}\text{C}$, as imposed, and the oil sump temperature in the REB chamber stabilizes at $62.7\text{ }^{\circ}\text{C}$ (approximately $5\text{ }^{\circ}\text{C}$ higher). However, at 2500 rpm and 3.0 Nm, the oil sump temperature in the Gear chamber stabilizes at $71.5\text{ }^{\circ}\text{C}$ instead of the imposed $68.5\text{ }^{\circ}\text{C}$ and the oil sump temperature in the REB chamber stabilizes at $78.3\text{ }^{\circ}\text{C}$ (approximately $10\text{ }^{\circ}\text{C}$ higher).

3.3.2. Torque loss measurements in the range 500–2500 rpm

The normal range of the operating speed of the axle box is between 500 and 2500 rpm, corresponding to vehicle speeds between 20 and 100 km/h.

Fig. 11 shows the influence of the input speed on the torque loss of the pinion assembly, for three different oil sump temperatures and three different starting torques (corresponding to three different rolling

bearing pre-loads). A constant immersion depth of $a_{\text{lub}} = 20\text{ mm}$ was considered, corresponding to the oil level recommended by the axle box manufacturer (see Fig. 4).

As expected, when the speed increases the torque loss also increases towards an almost asymptotic value at 2000 rpm. In addition, when the oil sump temperature increases, the oil viscosity decreases and the torque loss becomes smaller.

The influence of the starting torque (or pre-load) is also very significant. For an oil sump temperature of $62.2\text{ }^{\circ}\text{C}$ ($v = 42.9\text{ cSt}$) and an input speed of 2000 rpm, the torque loss of the pinion assembly increases from 1.35 Nm up to 1.60 Nm (+18%) when the starting torque is creased from 0.75 Nm to 1.90 Nm. When the starting torque was increased to 3.00 Nm the additional increase of the torque loss of the pinion assembly is smaller.

For low temperature and high speed (e.g. $T_{\text{Gear_sump}}^{\text{Gear}} = 50\text{ }^{\circ}\text{C}$ and $2000 \leq n \leq 2500\text{ rpm}$) the torque loss stabilizes or even decreases slightly. In fact, although the oil sump temperature is kept constant, locally (e.g. inside the rolling bearings), the mass temperature might rise above the oil sump temperature and consequently the viscosity drops and the torque loss decreases. The results were obtained considering the mass temperature of the bearing races is equal rolling bearing chamber temperature.

The results shown in Fig. 11 clearly indicate that the starting torque associated with the rolling bearings pre-load has a very significant influence on the pinion assembly torque loss.

3.4. Influence of the rolling bearings immersion depth

3.4.1. Temperature measurements at different immersion depths

Fig. 12 shows the stabilized temperatures in the REB and Gear chambers, for operating speeds in the range 500–2500 rpm, different immersion depths and a starting torque of 0.75 Nm.

Fig. 12 is very similar to Fig. 10 ($M_{\text{st}} = 0.75\text{ Nm}$) and shows the same trends. Fig. 12 also indicates that the temperature in the REB chamber decreases as the immersion depth decreases, which was expected.

3.4.2. Torque loss measurements at different immersion depths

The rolling bearings immersion depth influence on the torque loss is presented in Fig. 13, where the pinion assembly torque loss variation

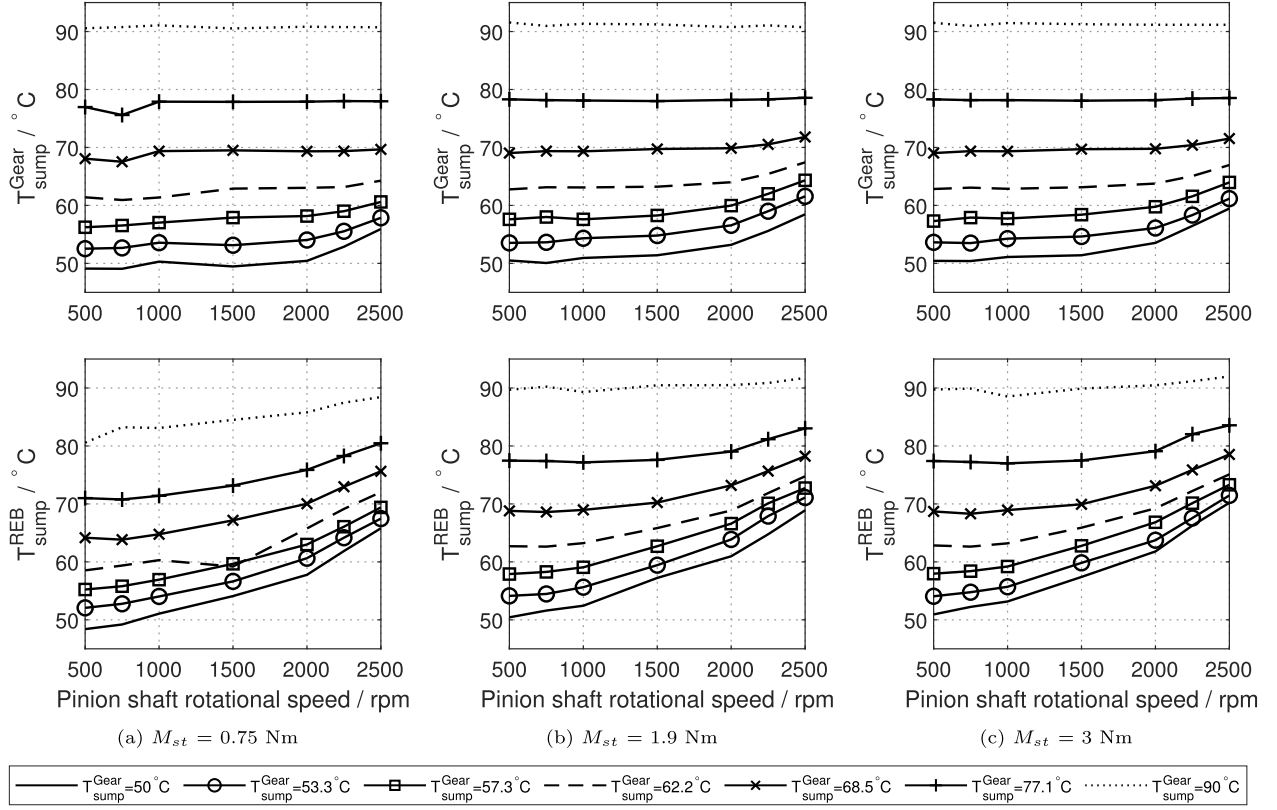


Fig. 10. Stabilized temperatures in the REB and Gear test chambers for different starting torques.

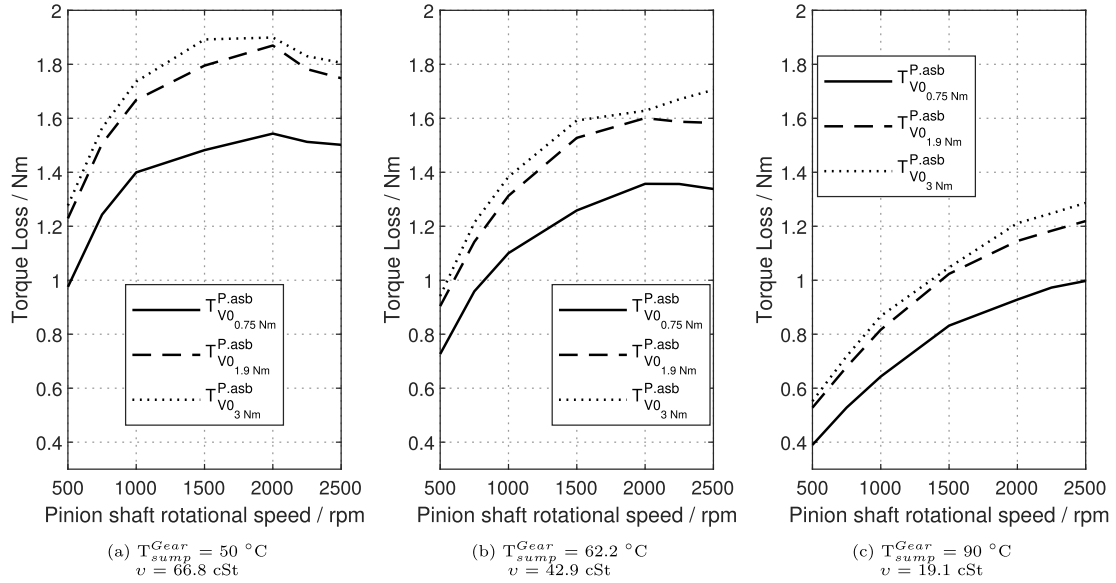


Fig. 11. Pinion assembly torque loss for different starting torques.

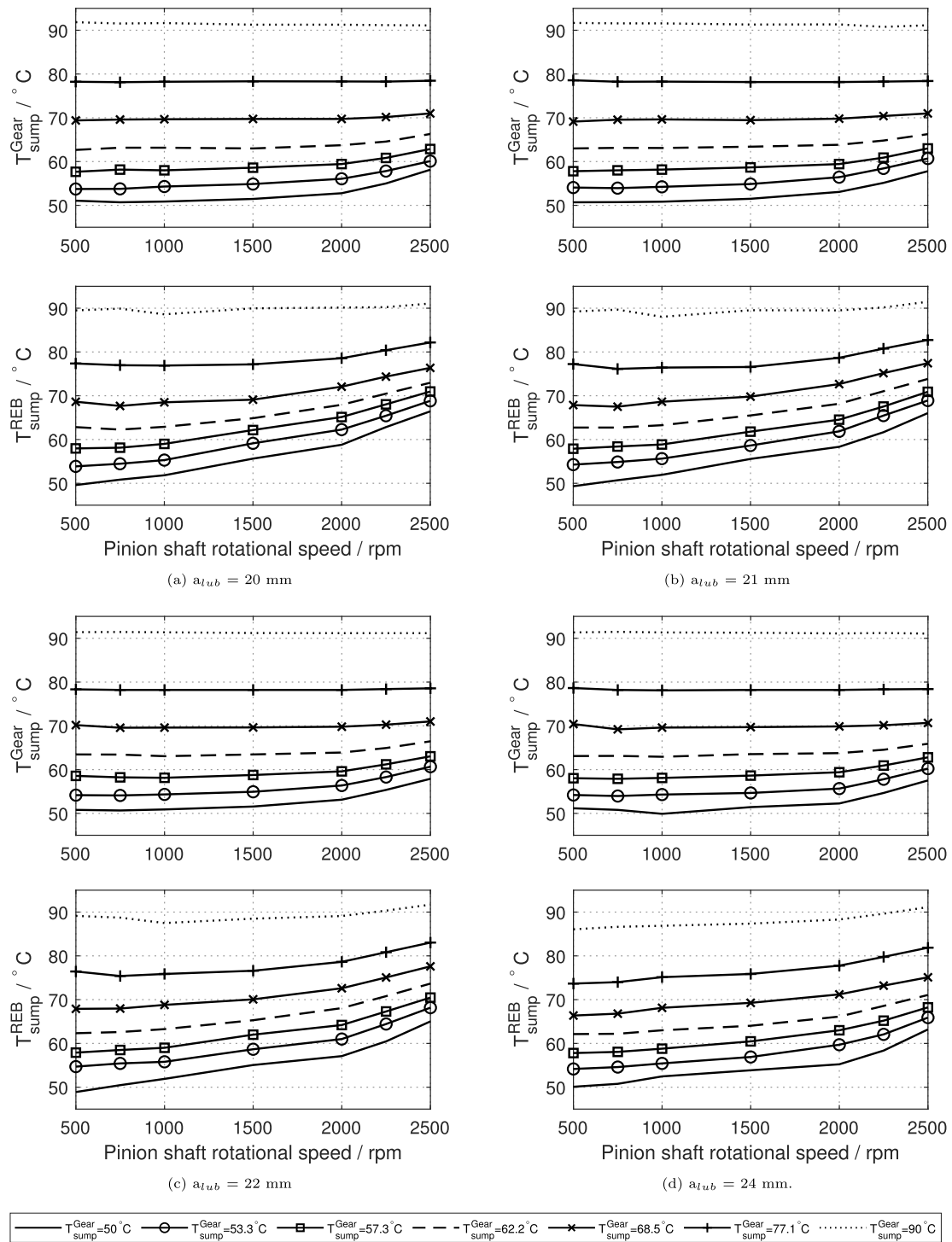


Fig. 12. Stabilized temperatures in the REB and Gear test chambers, for different immersion depth ($M_{st} = 0.75$ Nm).

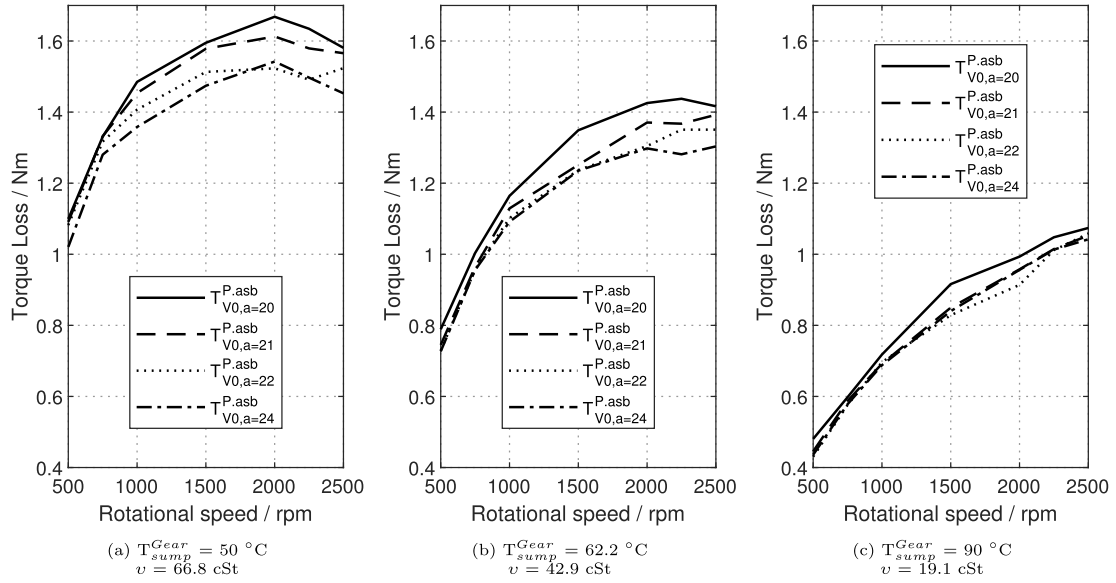


Fig. 13. Pinion assembly torque loss for different rolling bearings immersion depth.

with the rotational speed is shown, that includes four different immersion depths ($a_{lub} = 20, 21, 22$ and 24 mm) and three oil temperatures ($T_{sump}^{Gear} = 50^\circ, 62.2^\circ$ and 90°C).

A better understanding of the immersion depth relation with the rolling bearings can be visualized at Fig. 4.

As expected, generally a small torque loss decrease is verified with the a_{lub} increase.

4. Torque loss models for the pinion assembly

The scope of this section is to analyse the friction torque of the pinion assembly, according to equation (1).

4.1. Pinion churning torque loss model

The Fossier model [3,17,18], represented by equation (2), was used to estimate the pinion gear churning torque loss ($T_{VZ0,FS}^{P,g}$).

$$T_{VZ0,FS} = \frac{1}{2} \rho_{oil} \omega^2 r_o^3 S_m C_m^f \quad (2)$$

1. $Re_f \leq 20000$:

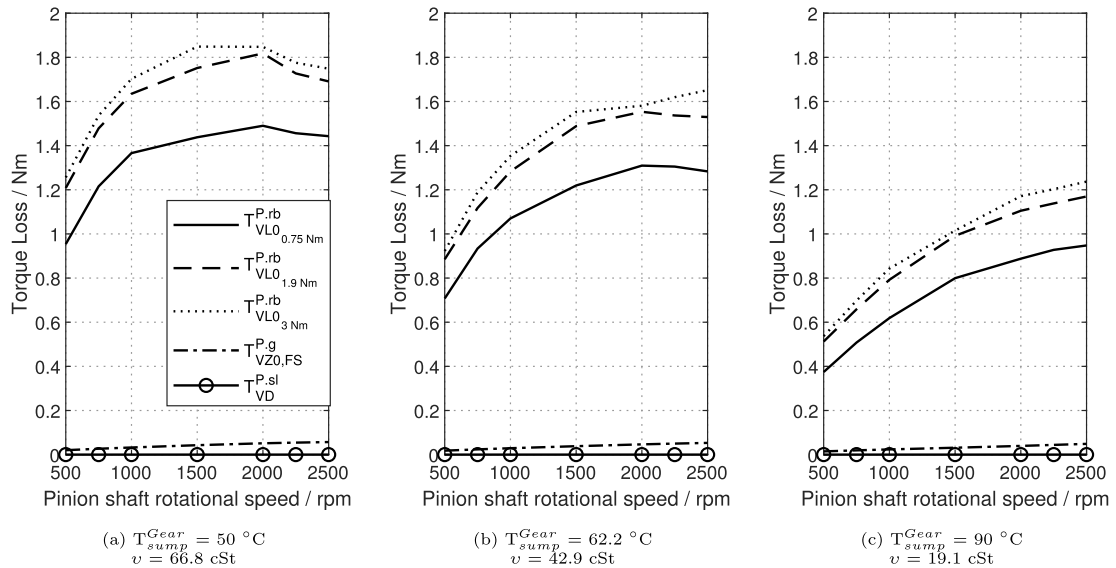
$$C_m^f = 1.45 \cdot \left(\frac{h}{r_o}\right)^{0.15} \cdot \left(\frac{V_0}{r_o^3}\right)^{-0.20} \cdot Re_f^{-0.25} \cdot Fr^{-0.53} \quad (3)$$

2. $Re_f > 20000$:

$$C_m^f = 1.12 \cdot \left(\frac{h}{r_o}\right)^{0.15} \cdot \left(\frac{V_0}{r_o^3}\right)^{-0.20} \cdot Fr^{-0.53} \quad (4)$$

The Reynolds and Froude numbers are defined according to equations (5) and (6).

$$Re_f = \frac{r_o^2 \omega}{\nu} \quad (5)$$

Fig. 14. Pinion rolling bearings torque loss for different starting torques ($a_{lub} = 20 \text{ mm}$).

$$Fr = \frac{r_o \cdot \omega^2}{g} \quad (6)$$

4.2. Pinion tapered rolling bearing torque loss model

The rolling bearings torque loss (T_{VLO}) can be estimated using the models proposed by the SKF [13], by the ISO standard [14] or by Witte [15], using equations (7) and (8) or (9), respectively.

$$T_{VLO,SKF} = M_{rr} + M_{sl} + M_{seal} + M_{drag} \quad (7)$$

$$T_{VLO,ISO} = M_0 + M_1 \quad (8)$$

$$M_{Witte} = 1.1 \cdot 10^{-4} \cdot G \cdot (n \cdot \mu)^{1/2} \cdot (F_a^*)^{1/3} \quad (9)$$

4.3. Seals torque loss model

The torque loss generated by the pinion shaft seal ($T_{VD}^{P,sl}$), can be predicted with reasonably accuracy by the Simrit [19] model, given by equation (10).

$$T_{VD,Simrit} = 7.34 \times 10^{-2} \cdot d_{sh}^2 \quad (10)$$

5. Rolling bearing torque loss

5.1. Experimental rolling bearing torque loss

In Fig. 14, the experimental results of the pinion shaft tapered roller bearings are represented. This torque loss was obtained by subtracting the pinion gear churning torque loss according to the Fossier [3] model and the pinion seal torque loss according to the Simrit [19] model to the pinion shaft torque loss, as expressed in equation (11).

$$\begin{aligned} T_{V0}^{P,asb} &= T_{V0}^{P,g} + T_{V0}^{P,rb} + T_{V0}^{P,sl} \\ T_{V0,Exp}^{P,rb} &= T_{V0}^{P,asb} - T_{V0}^{P,g} - T_{V0,FS}^{P,sl} - T_{VD,Simrit}^{P,sl} \end{aligned} \quad (11)$$

For the speed range considered, the values of $T_{V0}^{P,rb}$ are significantly higher than the values of $T_{V0}^{P,g}$ and $T_{V0}^{P,sl}$, due to rolling bearings pre-load which is required to assure the high rigidity of the pinion assembly.

5.2. Starting torque

The starting torque is the sum of the starting torques resulting from each one of the tapered roller bearings, that is,

$$M_{st} = M_{st1} + M_{st2} \quad (12)$$

According to the SKF [13] rolling bearing friction torque model, at the onset of movement, the starting torque corresponds to the sliding friction torque causing boundary lubricant conditions ($M_{st} = M_{sl}$), since the remaining rolling bearing friction torque components (M_{rr} , rolling torque and M_{drag} , drag torque), which are dependent on the rolling bearing speed, are negligible.

The relations between the starting torque and the axial preload in each tapered roller bearing are established by the following expressions:

$$M_{st1} = M_{sl1} = \mu_{bl} \cdot S_1 \cdot d_{m1}^{0.82} \cdot S_2 \cdot Y \cdot F_a \quad (13)$$

And

$$M_{st2} = M_{sl2} = \mu_{bl} \cdot S_2 \cdot d_{m2}^{0.82} \cdot S_2 \cdot Y \cdot F_a \quad (14)$$

Since the total starting torque is measured and known, the geometric constants S1 and S2 are known from the SKF catalogue, and the mean diameter of each tapered roller bearing is known, it is possible to determine the corresponding axial preload, F_a .

Using the presented equation system (15).

$$\begin{cases} F_a \simeq \frac{M_{st1}}{\mu_{bl} \cdot S_1 \cdot d_{m1}^{0.82} \cdot S_2 \cdot Y} \\ F_a \simeq \frac{M_{st2}}{\mu_{bl} \cdot S_1 \cdot d_{m2}^{0.82} \cdot S_2 \cdot Y} \\ M_{st} = M_{st1} + M_{st2} \end{cases} \quad (15)$$

Using the system of equation (15), the following preloads were estimated: 2083 N at 0.75 Nm, 5279 N at 1.9 Nm and 8336 N at 3 Nm. These preloads are necessary to estimate the roller bearings torque loss using SKF [13], ISO [14] and Witte [15] models.

5.3. Experimental vs models RB torque loss

The pinion rolling bearings torque loss was estimated using the SKF [13], ISO [14] and Witte [15] models and compared to the experimental

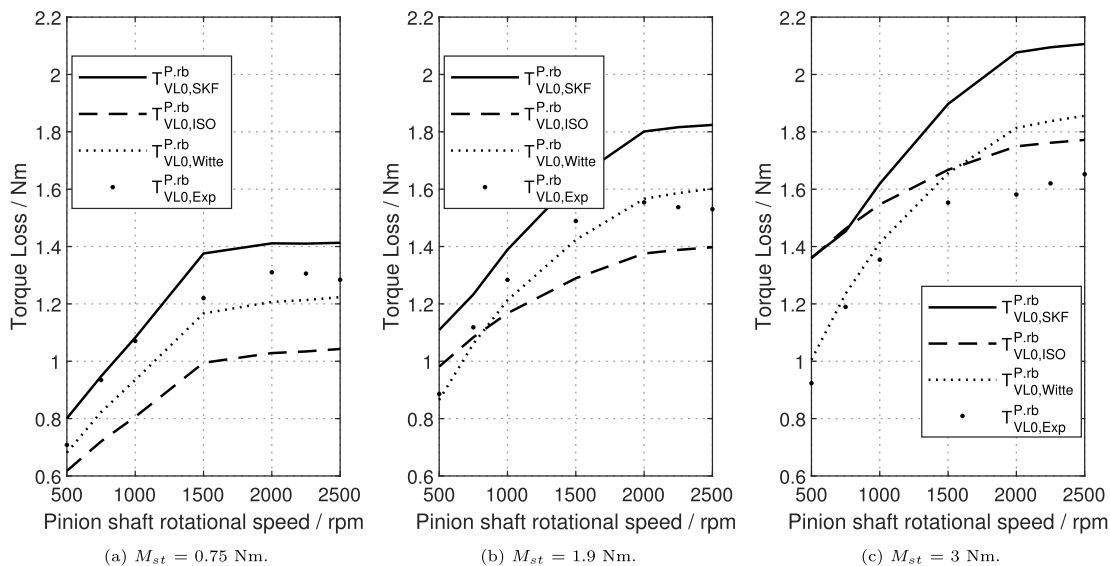


Fig. 15. Rolling bearing models comparison at $v = 42.9$ cSt @ $T_{sump}^{Gear} = 62.2$ °C.

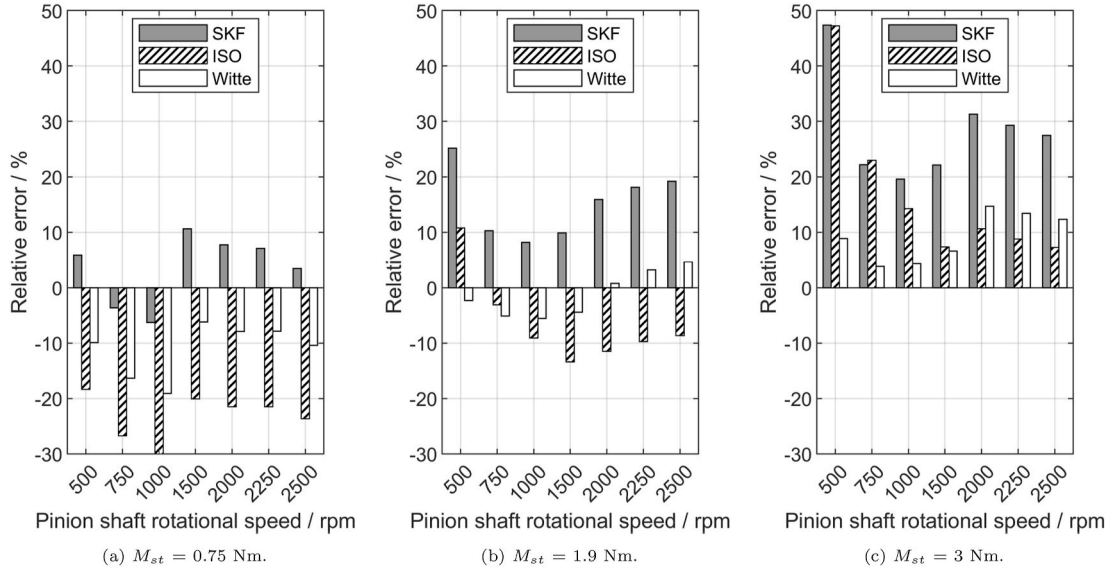


Fig. 16. Rolling bearing models relative error at $v = 42.9$ cSt @ $T_{sump}^{Gear} = 62.2$ °C.

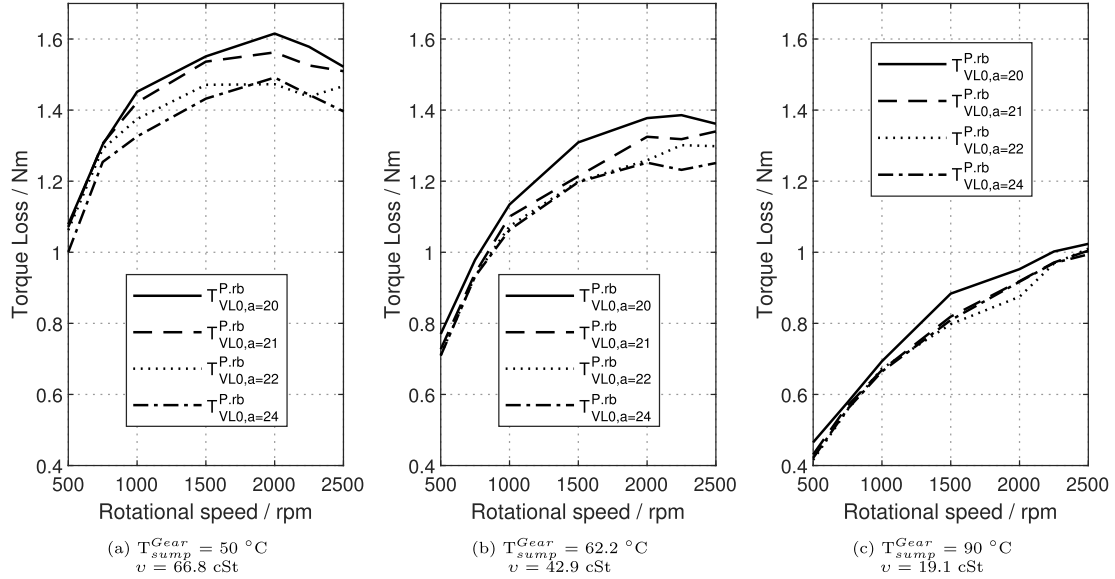


Fig. 17. Pinion rolling bearings torque loss for different immersion depths.

results (Fig. 15). The relative error between the experimental results can be observed in Fig. 16.

The sensitivity of the ISO [14] and SKF [13] to the starting torque is quite different. The SKF [13] model presented better estimations at the lowest preload ($M_{st} = 0.75$ Nm), overestimating the torque loss at higher starting torques. This difference is most likely to be induced by the friction coefficient in boundary film conditions ($\mu_{bl} = 0.15$ when $n = 0$) used to obtain the preloads. The model proposed by Witte, using the estimated axial load (equation (15)) is usually below the experimental measurements, but much closer to the real values than the ISO model. It could be said that for $M_{st} = 1.9$ Nm and $M_{st} = 3.0$ Nm, Witte's model is actually closer to reality than any of the other two models. This may not be surprising, since this model was developed for axially loaded tapered roller bearings.

5.4. Influence of the immersion depth

Similar to the procedure presented in section 5.1, the pinion gear churning torque loss according to the Fossier [3] model and the pinion seal torque loss according to the Simrit [19] model were subtracted at the pinion assembly torque loss results for the immersion depth influence analysis, as expressed in equation (11), obtaining the results presented at Fig. 17.

As it was expected, a decrease of the rolling bearings torque loss is verified with the a_{lub} increase, explained by the drag losses decrease, although at the higher a_{lub} ($a_{lub} = 24$ mm), the tendency inversion can be verified at some points, explained by the smaller rolling bearing starvation at this oil level.

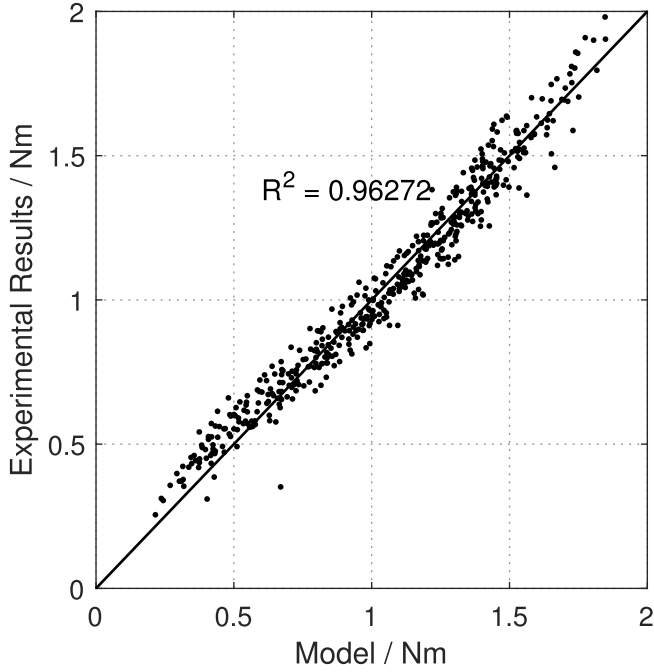


Fig. 18. Equation comparison with experimental results.

6. Tapered roller bearings torque loss

No model was found in the literature that could give estimations of the torque loss for tapered rolling bearing assembly that matched the measured values.

6.1. An empiric equation for the tapered R.B. torque loss

As thoroughly shown in Appendix A, the tapered roller bearings torque loss is the sum of two components, the transient torque loss and the hydrodynamic torque loss, both depending on the starting torque generated by the pre-load applied to the tapered roller bearing assembly. The transient torque loss is given by equation (16). It depends on the starting torque and decreases very rapidly as the speed increases, as shown in Fig. 9.

$$M_{transient} = M_{st} \cdot e^{-\frac{\omega}{\omega_0}} \quad (16)$$

The hydrodynamic torque is given by equation (17). It also depends on the starting torque, as shown by the experimental measurements shown on Figs. 8 and 14, and on the dynamic viscosity of the oil, on the operating speed, on the Reynolds number and on the taper rolling bearings dimensions (dm and T). The Reynolds number is used, so the viscous and inertial behaviours of the lubricant are also considered, that is included in the dimensionless coefficient Cm , with the oil level high and the rolling bearing mean radius.

$$M_{hydrodynamic} = (M_{st} + C_1)^\alpha \cdot \left[(\eta \cdot \omega \cdot dm_1^2 \cdot T_1)^{1-\alpha} \cdot Cm_1 + (\eta \cdot \omega \cdot dm_2^2 \cdot T_2)^{1-\alpha} \cdot Cm_2 \right] \quad (17)$$

where the dimensionless constant Cm_i is given by equation (18).

$$Cm_i = C_2 \cdot \left(\frac{h}{R_m} \right)^\beta \cdot Re_r^\theta, \quad i = 1, 2 \quad (18)$$

The values of the exponents α , β and θ , as well as the values of the constants C_1 , C_2 and ω_0 were determined by correlation with the experimental measurements: $\alpha = 0.45$, $\beta = 0.28$, $\theta = -0.080$, $C_1 = 1.86$, $C_2 = 56.3$ and $\omega_0 = 2 \text{ rad s}^{-1}$. The Reynolds number is defined according equation (19).

$$Re_r = \frac{R_m^2 \cdot \omega}{\nu} \quad (19)$$

Equation (20) shows the torque loss of the taper roller bearing assembly. A comparison between equation (20) and the experimental results is shown in Fig. 18, where a correlation factor of 0.963 was found.

$$T_{VL0} = M_{st} \cdot e^{-\frac{\omega}{\omega_0}} + (M_{st} + C_1)^\alpha \cdot \left[(\eta \cdot \omega \cdot dm_1^2 \cdot T_1)^{1-\alpha} \cdot Cm_1 + (\eta \cdot \omega \cdot dm_2^2 \cdot T_2)^{1-\alpha} \cdot Cm_2 \right] \quad (20)$$

Fig. 19 compares the experimental measurements and the new equation estimations for the torque loss of the tapered roller bearings assembly for an oil sump temperature of 77.1°C ($\nu = 27.7 \text{ cSt}$) and for the three starting torques tested. Similarly good correlations were obtained for all the other operating temperatures, from 50°C up to 90°C .

For the moment, equation (20) has only been verified for the tandem tapered roller bearings of the axle box (ref.33107518807) shown in Table 2.

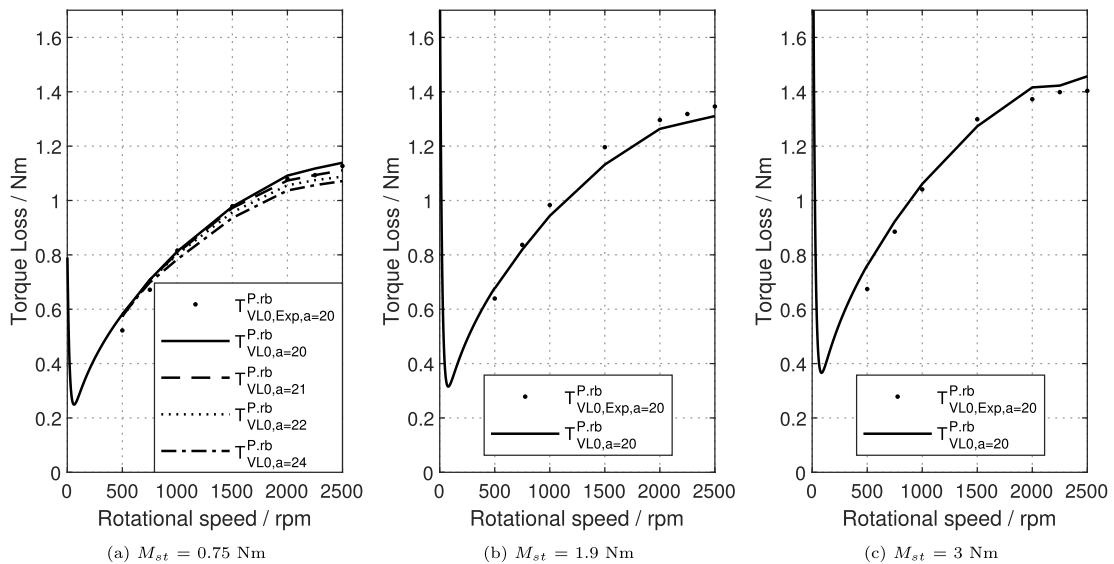


Fig. 19. Rolling bearing torque loss equation comparison with the experimental results at $\nu = 27.0 \text{ cSt}$ @ $T_{Gear\ sump} = 77.1^\circ\text{C}$.

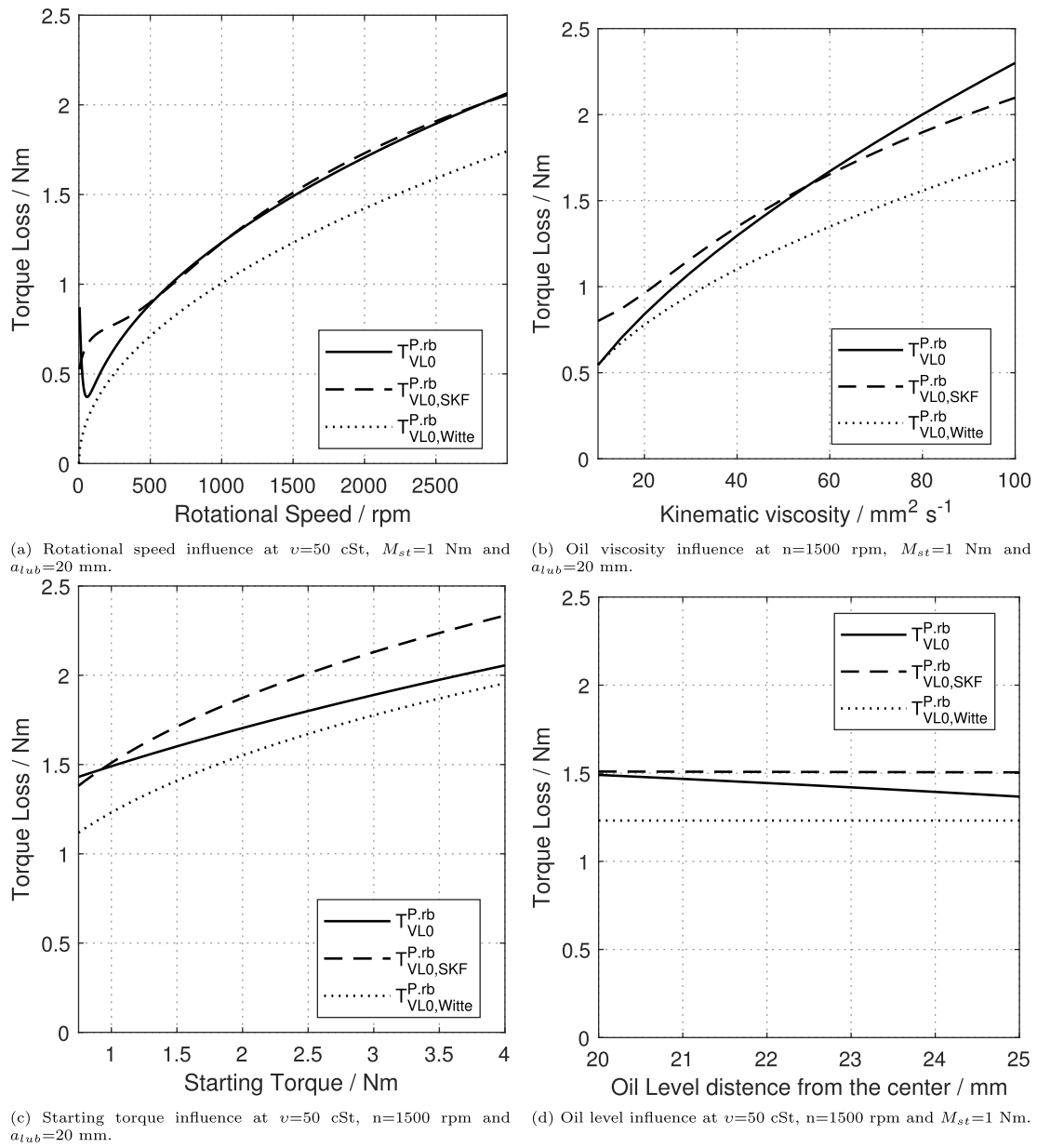


Fig. 20. Equation sensibility analysis.

6.2. Equation sensitivity analysis

The equation has shown good estimations of the rolling bearings torque loss when compared to the experimental results. A sensibility analysis was also performed, represented in Fig. 20, which studies the influence of each equation variable and compare the tendency with the SKF [13] and Witte [15] models. The simulations were done by considering: $\nu = 50$ cSt, $n = 1500$ rpm, $M_{st} = 1$ Nm and $a_{lub} = 20$ mm.

The results shown in Fig. 20 show that the presented equation reacts to parameters such as speed, viscosity, oil level and starting torque similarly to SKF and Witter's model. There is an agreement between the presented equation and existing state of the art models in terms of sensitivity to there key parameters.

7. Conclusions

The importance of the vehicle efficiency is increasing, being the rolling bearings one of main aspects that can be improved. So, in this work, the tapered roller bearings of a rear axle were studied, where the influence of pre-load, rotational speed, lubricant viscosity and oil level were analysed in the friction torque. This study allowed to better understand the power loss in the rolling bearings and to compare the estimations presented by some models found in the literature.

The experimental results allowed to conclude that a starting torque increase leads to a torque loss increase, until a certain value, where it remains constant, effect explained by the pre-load increase and the hydrodynamic behaviour at higher speeds. As expected, the lubricant viscosity and rotational speed increase led to a torque loss increase. A high torque loss is verified at the starting of the movement of the rolling bearings, that decreases very rapidly down to a value where it increases again. The lubricant level decrease induces a torque loss decrease down to a certain point, where the bearings lubricant starvation starts to be verified.

The experimental results allowed to perform a comparison between the main models presented in the literature, where some discrepancies where verified, mainly due to models reaction to the operating conditions variation. The Witte model has shown the better estimations when compared with the experimental results, although any model has shown

a suitable reaction to the pre-load increase.

An empirical equation capable of estimating the rolling bearings torque loss was developed, based on the experimental results, where instead of the preload, the starting torque was considered. The developed equation is based on a theoretical model, that was developed and explained.

To conclude, the developed equation was analysed, where a good correlation with the experimental results was obtained. To assure a good behaviour of the presented equation, a sensibility analysis was performed, where the presented equation reacted similarly to the SKF and Witte models.

CRedit authorship contribution statement

Justino A.O. Cruz: Conceptualization, Methodology, Data curation, Investigation, Writing - original draft. **Pedro M.T. Marques:** Conceptualization, Validation, Writing - review & editing, Supervision. **Jorge H.O. Seabra:** Conceptualization, Resources, Writing - review & editing, Visualization, Project administration. **Ramiro C. Martins:** Formal analysis, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the funding whom without this work would not have been possible:

- FCT PhD programme NORTE-69-2015-15 - supported by Norte Portugal Regional Operational Programme (NORTE 2020), under the PORTUGAL 2020 Partnership Agreement, through the European Social Fund (ESF) having the funding received the reference NORTE-08-5369-FSE-000027;
- LAETA under the project UID/50022/2020.

Appendix A. Tapered roller bearing (TRB) torque loss - Houpert's model

Houpert [20] developed a theoretical model for the torque loss in tapered roller bearings, represented by equations (A.1) and (A.2).

$$dT = dT_{FR} + dT_{rib} + dT_{MER} \quad (A.1)$$

$$dT = 2 \cdot (FR_o + FR_i) \cdot \frac{R_o \cdot R_i}{d_m} + \frac{Mrib \cdot R_o}{d} + \frac{MER_i \cdot R_o + MER_o \cdot R_i}{d} \quad (A.2)$$

This equation gives the torque acting on each rolling element, which depends on the hydrodynamic resistant forces (FR) in the inner and outer raceways, on the elastic rolling moment (MER) and on the RIB torque (M_{rib}).

Equation (A.3) is derived from the previous equation by replacing the outer (R_o) and inner (R_i) race radius of the bearing by some approximations based on the bearing mean diameter and the constant γ . So, the rolling element torque is expressed for each element in classical parameters (rolling bearing geometrical parameters, axial load and friction coefficients), considering only the inner ring rotation,

$$dT = 0.5 \cdot dm \cdot (1 + \gamma) \cdot (1 - \gamma) \cdot (FR_o + FR_i) + 0.212 \cdot Z \cdot ((1 + \gamma)(Mrib + MER_i) + (1 - \gamma) \cdot MER_o) \quad (A.3)$$

$$\text{where } \gamma = \frac{d \cdot \cos \alpha^*}{d_m}.$$

Appendix A.1. Hydrodynamic Race Torque

Using the displacement of pressure center expression, presented by Hamrock [21], Houpert [22] derived the rolling traction force for a line contact under full-film EHD lubrication, given by equation (A.4),

$$FR = 0.04 \cdot E' \cdot R_x \cdot l \cdot U^{0.44} \cdot W^{0.37} \quad (A.4)$$

where E' is the equivalent Young's modulus, R_x is the equivalent radius in the rolling direction, l is the roller length, U is a dimensionless speed parameter and W is a dimensionless load.

The rolling traction force expression can be expressed in classical parameters, as presented in equations (A.5) and (A.6), for the inner race (i) and outer race (o) contacts, respectively.

$$FR_i = 0.0165 \cdot E'^{0.19} \cdot Z^{-0.56} \cdot dm^{0.63} \cdot l^{0.63} \cdot (1 + \gamma)^{-0.19} \cdot (\eta \cdot \omega_i \cdot (1 - \gamma^2))^{0.44} \cdot (F_a \cdot \sin \alpha^*)^{0.37} \quad (A.5)$$

$$FR_o = 0.0165 \cdot E'^{0.19} \cdot Z^{-0.56} \cdot dm^{0.63} \cdot l^{0.63} \cdot (1 - \gamma)^{-0.19} \cdot (\eta \cdot \omega_i \cdot (1 - \gamma^2))^{0.44} \cdot (F_a \cdot \sin \alpha^*)^{0.37} \quad (A.6)$$

where the roller diameter and the normal force approximations were considered, as expressed in equations (A.7) and (A.8).

$$d \approx \frac{\pi \cdot dm}{1.33 \cdot Z} \quad (A.7)$$

$$Q \approx \frac{F_a \cdot \sin \alpha^*}{Z} \quad (A.8)$$

The rolling traction force can be converted in torque using equation (A.3), obtaining equation (A.9).

$$dT_{FR} = 0.5 \cdot dm \cdot (1 + \gamma) \cdot (1 - \gamma) \cdot (FR_o + FR_i) \quad (A.9)$$

$$\begin{aligned} &= 0.00825 \cdot (1 - \gamma) \cdot (1 + \gamma) \cdot (1 - \gamma^2)^{0.44} \cdot ((1 - \gamma)^{-0.19} + (1 + \gamma)^{-0.19}) \cdot Z^{-0.56} \cdot \sin \alpha^* \\ &\cdot E'^{0.19} \cdot dm^{1.63} \cdot l^{0.63} \cdot (\eta \cdot \omega_i)^{0.44} \cdot F_a^{0.37} \\ &= 0.00825 \cdot \gamma^* \cdot \sin \alpha^* \cdot Z^{-0.56} \cdot E'^{0.19} \cdot dm^{1.63} \cdot l^{0.63} \cdot (\eta \cdot \omega_i)^{0.44} \cdot F_a^{0.37} \end{aligned}$$

where γ^* is given by equation (A.10).

$$\gamma^* = (1 - \gamma) \cdot (1 + \gamma) \cdot (1 - \gamma^2)^{0.44} \cdot [(1 - \gamma)^{-0.19} + (1 + \gamma)^{-0.19}] \quad (A.10)$$

Appendix A.2. RIB Torque

The contact between each roller and the rib results in a moment, expressed in (A.11), characteristic of tapered roller bearings, which depends only on the rolling bearing geometry.

$$M_{rib} \approx \mu_{rib} \cdot Q_{rib} \cdot f \cdot \sqrt{1 + 0.18 \left(\frac{b}{f} \right)^2} \quad (A.11)$$

Similar to the rolling traction force, the M_{rib} can be expressed in classical rolling bearings parameters, as given by equation A.12.

$$M_{rib} \approx 2 \cdot \mu_{rib} \left(\frac{F_a \cdot \sin \alpha^*}{Z} \right) \sin \nu \cdot f \cdot \sqrt{1 + 0.18 \left(\frac{b}{f} \right)^2} \quad (A.12)$$

where ν is half the included roller angle, of the order of 2° , f is the height of the roller - rib contact, b is half of the rolling element - race contact and μ_{rib} is the local friction coefficient.

After applying the ratio between dimensions, as expressed in equation (A.3), equation (A.13) is obtained, where the geometric parameters were replaced by the variable B and divided in both the transient and the EHD regime, using a weighting factor ϕ_{bl} similar to the one presented by the SKF [13].

$$\begin{aligned} dT_{rib} &= 0.212 \cdot Z \cdot (1 + \gamma) \cdot M_{rib} = 0.424 \cdot (1 + \gamma) \cdot \sqrt{1 + 0.18 \left(\frac{b}{f} \right)^2} \cdot \sin \nu \cdot \sin \alpha^* \cdot \mu_{rib} \cdot F_a \cdot f = 0.424 \cdot B \cdot \sin \alpha^* \cdot \mu_{rib} \cdot f \cdot F_a \\ &= 0.424 \cdot B \cdot \phi_{rib} \cdot f \cdot \sin \alpha^* \cdot \mu_{bl} \cdot F_a + 0.424 \cdot B \cdot (1 - \phi_{rib}) \cdot f \cdot \sin \alpha^* \cdot \mu_{EHD} \cdot F_a \end{aligned} \quad (A.13)$$

where the variable B and the RIB friction coefficient μ_{rib} are given by equations (A.14) and (A.15), respectively.

$$B = \left(1 + \gamma \right) \cdot \sin \nu \cdot \sqrt{1 + 0.18 \left(\frac{b}{f} \right)^2} \quad (A.14)$$

$$\mu_{rib} = \phi_{rib} \cdot \mu_{bl} + (1 - \phi_{rib}) \cdot \mu_{EHD} \quad (A.15)$$

Appendix A.3. Elastic Rolling Moment

The elastic rolling moment (MER) is attributed to the hysteresis losses and can be represented by equation (A.16).

$$MER = 0.00042 \cdot Q \cdot R_x \cdot \frac{\mu_e}{0.11} \quad (\text{A.16})$$

When expressed in classical parameters, the equations (A.17) and (A.18) are obtained,

$$MER_i = 0.004509 \cdot \frac{\sin \alpha^*}{Z^2} \cdot F_a \cdot dm \cdot (1 + \gamma)^{-1} \cdot \mu_e \quad (\text{A.17})$$

$$MER_o = 0.004509 \cdot \frac{\sin \alpha^*}{Z^2} \cdot F_a \cdot dm \cdot (1 - \gamma)^{-1} \cdot \mu_e \quad (\text{A.18})$$

Where μ_e is the equivalent friction coefficient.

According to equation (A.3), the final elastic rolling moment is calculated using equation (A.19).

$$\begin{aligned} dT_{MER} &= 0.212 \cdot Z \cdot ((1 + \gamma) \cdot MER_i + (1 - \gamma) \cdot MER_o) \\ &= 0.001912 \cdot \frac{\sin \alpha^*}{Z} \cdot F_a \cdot dm \cdot \mu_e \\ &= 0.001912 \cdot \frac{\sin \alpha^*}{Z} \cdot F_a \cdot dm \cdot \phi_{mer} \cdot \mu_{bl} \end{aligned} \quad (\text{A.19})$$

Where $\mu_e = \phi_{mer} \cdot \mu_{bl}$.

Appendix A.4. Model torque loss prediction

Replacing equations (A.9), (A.13) and (A.19) in equation (A.1), the torque acting on each rolling element of a tapered roller bearing (TRB) becomes,

$$dT = dT_{FR} + dT_{rib} + dT_{MER} \quad (\text{A.20})$$

$$\begin{aligned} dT &= 0.00825 \cdot \gamma^* \cdot \sin \alpha^* \cdot Z^{-0.56} \cdot E^{0.19} \cdot dm^{1.63} \cdot f^{0.63} \cdot (\eta \cdot \omega_i)^{0.44} \cdot F_a^{0.37} \\ &+ 0.424 \cdot B \cdot f \cdot \sin \alpha^* \cdot \phi_{rib} \cdot \mu_{bl} \cdot F_a \\ &+ 0.424 \cdot B \cdot f \cdot \sin \alpha^* \cdot (1 - \phi_{rib}) \cdot \mu_{EHD} \cdot F_a \\ &+ 0.001912 \cdot \frac{\sin \alpha^*}{Z} \cdot dm \cdot \phi_{mer} \cdot \mu_{bl} \cdot F_a \end{aligned}$$

Equation (A.20) clearly shows that the torque acting on each roller element depends on the TRB geometry (Z , γ^* , α^* , l , d_m , b , f), on the lubricant dynamic viscosity (η), on the inner raceway speed (ω_i), on the axial load (F_a) and on the coefficient of friction (ϕ_{rib} , ϕ_{mer} , μ_{bl} , μ_{EHD}).

Equation (A.20) was applied to the case of the TRB HM 801349/310 (see Table 2), one of the TRB supporting the pinion shaft of the differential. Figure A.21a shows the different components of the torque on each rolling element of that TRB, considering an oil viscosity of $\nu = 27.7$ cSt ($\eta = 0.0240$ Pa s @ 77.1°C) and an axial load of 2083 N (see section 5.2).

Regarding the roller element torque loss components, the transient and the hydrodynamic regimes can be easily distinguished, as the transient regimes tend to start at a defined torque loss and progressively decrease to zero as the speed increases, while the hydrodynamic losses start from zero and increase continuously with the rotational speed.

Multiplying equation (A.20) by the number of rollers of the TRB, equation (A.21) is obtained, giving the torque loss on the TRB,

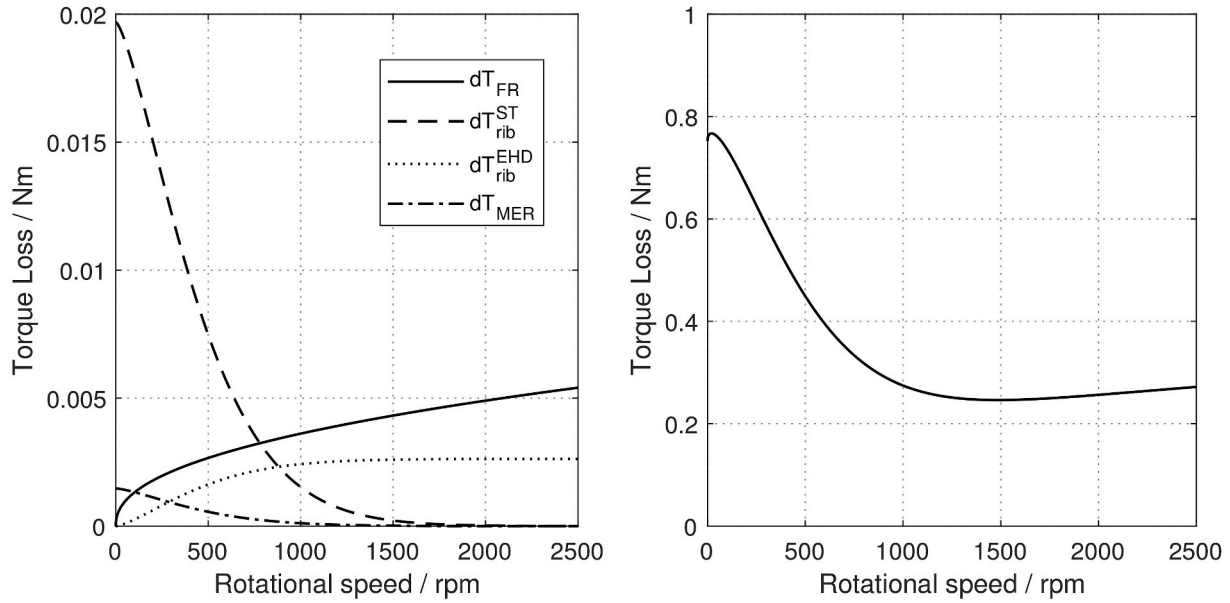
$$M_i = dT \cdot Z \quad (\text{A.21})$$

$$\begin{aligned} M &= 0.00825 \cdot \gamma^* \cdot \sin \alpha^* \cdot Z^{0.44} \cdot E^{0.19} \cdot dm^{1.63} \cdot f^{0.63} \cdot (\eta \cdot \omega_i)^{0.44} \cdot F_a^{0.37} \\ &+ 0.424 \cdot B \cdot f \cdot Z \cdot \sin \alpha^* \cdot \phi_{rib} \cdot \mu_{bl} \cdot F_a \\ &+ 0.424 \cdot B \cdot f \cdot Z \cdot \sin \alpha^* \cdot (1 - \phi_{rib}) \cdot \mu_{EHD} \cdot F_a \\ &+ 0.001912 \cdot \sin \alpha^* \cdot dm \cdot \phi_{mer} \cdot \mu_{bl} \cdot F_a \end{aligned}$$

The torque loss of TRB expressed by equation (A.21) shows the importance of the starting torque (generated by the axial pre-load applied to the tandem roller bearings, F_a), present in both the hydrodynamic and the transient regimes. In fact, the torque loss on each tapered roller bearing has four components: two of them depend on the axial pre-load (F_a) and on the hydrodynamic component ($\omega \times \eta$) and the remaining two only depend on the axial pre-load, and thus on the starting torque.

Equation (A.21) was applied to the TRB's supporting the pinion shaft of the differential (see Table 2) in a tandem assembly, considering the

geometry of each rolling bearing. Figure A.21b presents the prediction of the torque loss generated by the tandem TRB, for different speeds of the inner race, which shows a trend similar to the experimental measurements presented in Figs. 8 and 14. A lower influence of the rotational speed is observed.



(a) Components of the torque applied to each roller of the HM 801349/310 TRB. (b) Torque loss in the tandem TRB supporting the differential pinion shaft.

Fig. A.21. Tapered roller bearing torque loss according to Houpert's model [20].[]

Appendix A.5. Starting torque

According to the formulation presented by Houpert [20], the starting torque can be expressed by the RIB torque and the elastic rolling moment at the onset of movement, where ϕ_{mer} and ϕ_{rib} assume a unitary value ($\phi_{mer}=1$ and $\phi_{rib}=1$).

For the tandem TRB's, equation (A.21) becomes

$$M_{st} = M_{st1} + M_{st2} \quad (A.22)$$

$$= [(0.424 \cdot B_1 \cdot f_1 \cdot Z_1 + 0.001912 \cdot dm_1) \cdot \sin \alpha_1^* + (0.424 \cdot B_2 \cdot f_2 \cdot Z_2 + 0.001912 \cdot dm_2) \cdot \sin \alpha_2^*] \cdot \mu_{bl} \cdot F_a$$

$$= \left[\left(0.424 \cdot \frac{B_1 \cdot f_1 \cdot Z_1}{dm_1} + 0.001912 \right) \cdot \sin \alpha_1^* \cdot dm_1 + \left(0.424 \cdot \frac{B_2 \cdot f_2 \cdot Z_2}{dm_2} + 0.001912 \right) \cdot \sin \alpha_2^* \cdot dm_2 \right] \cdot \mu_{bl} \cdot F_a$$

$$M_{st} = (K_1 \cdot \sin \alpha_1^* \cdot dm_1 + K_2 \cdot \sin \alpha_2^* \cdot dm_2) \cdot \mu_{bl} \cdot F_a \quad (A.23)$$

The axial load can now be obtained as expressed in equation (A.24).

$$F_a = \frac{M_{st}}{(K_1 \cdot \sin \alpha_1^* \cdot dm_1 + K_2 \cdot \sin \alpha_2^* \cdot dm_2) \cdot \mu_{bl}} \quad (A.24)$$

where K is expressed by equation (A.25),

$$K_i = (K_i' + D) \quad K_i' = C \cdot \frac{B_i \cdot f_i \cdot Z_i}{dm_i} \quad i = 1, 2 \quad (A.25)$$

Equation (A.23) is very interesting, since it shows that the starting torque depends on the axial load and on the mean diameter, as expected, but it also depends on the geometry (B_i , t_i , Z_i , α_i^*) of the two TRB in a tandem assembly.

Thus, the measured starting torque takes into account a global information about axial load, geometry and friction (μ_{bl}) of the tandem assembly.

In section 5.2, the relation between the starting torque and the axial load of the tandem TRB's was established. The correlation between the measured starting torques and axial loads and the prediction of equations (A.23) and (A.25) allows the definition of the constants C and D necessary to define K_1 and K_2 in equation (A.25), which are $C = 0.943$ and $D = 0.004252$, respectively, that is

$$K_i = K_i' + 0.004252 \quad (i = 1, 2) \quad (A.26)$$

$$K_i' = 0.943 \cdot \frac{B_i \cdot f_i \cdot Z_i}{dm_i} \quad (i = 1, 2) \quad (A.27)$$

Appendix A.6. Empirical model for the torque loss of tandem TRB's

Summing the torque loss of the both tandem TRB's, equation (A.21) can now be written in its final form, assuming a global weighting factor ϕ ($\phi \approx \phi_{RIB} \approx \phi_{mer}$).

$$M = M_1 + M_2 \quad (A.28)$$

$$\begin{aligned} &= 0.00825 \cdot (\gamma_1^* \cdot \sin \alpha_1^* \cdot Z_1^{0.44} \cdot dm_1^{1.63} \cdot l_1^{0.63} + \gamma_2^* \cdot \sin \alpha_2^* \cdot Z_2^{0.44} \cdot dm_2^{1.63} \cdot l_2^{0.63}) \cdot E^{0.19} \cdot (\eta \cdot \omega_i)^{0.44} \cdot F_a^{0.37} \\ &+ (K'_1 \cdot \sin \alpha_1^* \cdot dm_1 + K'_2 \cdot \sin \alpha_2^* \cdot dm_2) \cdot (1 - \phi) \mu_{EHD} \cdot F_a \\ &(K_1 \cdot \sin \alpha_1^* \cdot dm_1 + K_2 \cdot \sin \alpha_2^* \cdot dm_2) \cdot \phi \cdot \mu_{bl} \cdot F_a \end{aligned}$$

Since the starting torque and the axial load are related by equation (A.24), equation (A.28) finally becomes

$$M = M_{transient}^* + M_{hydrodynamic}^* \quad (A.29)$$

where

$$M_{transient}^* = (K_1 \cdot \sin \alpha_1^* \cdot dm_1 + K_2 \cdot \sin \alpha_2^* \cdot dm_2) \cdot \phi \cdot \mu_{bl} \cdot F_a \quad (A.30)$$

$$M_{transient}^* = \phi \cdot M_{st} \quad (A.31)$$

$$M_{hydrodynamic}^* = (K'_1 \cdot \sin \alpha_1^* \cdot dm_1 + K'_2 \cdot \sin \alpha_2^* \cdot dm_2) \cdot (1 - \phi) \cdot \mu_{EHD} \cdot F_a + \quad (A.32)$$

$$\begin{aligned} &+ 0.00825 \cdot (\gamma_1^* \cdot \sin \alpha_1^* \cdot Z_1^{0.44} \cdot dm_1^{1.63} \cdot l_1^{0.63} + \gamma_2^* \cdot \sin \alpha_2^* \cdot Z_2^{0.44} \cdot dm_2^{1.63} \cdot l_2^{0.63}) \\ &\cdot E^{0.19} \cdot (\eta \cdot \omega_i)^{0.44} \cdot F_a^{0.37} \end{aligned}$$

$$M_{hydrodynamic}^* = (1 - \phi) \cdot \frac{\mu_{EHD}}{\mu_{bl}} \cdot \frac{K'_1 \cdot \sin \alpha_1^* \cdot dm_1 + K'_2 \cdot \sin \alpha_2^* \cdot dm_2}{K_1 \cdot \sin \alpha_1^* \cdot dm_1 + K_2 \cdot \sin \alpha_2^* \cdot dm_2} \cdot M_{st} + \quad (A.33)$$

$$\begin{aligned} &+ 0.00825 \cdot \frac{\gamma_1^* \cdot \sin \alpha_1^* \cdot Z_1^{0.44} \cdot dm_1^{1.63} \cdot l_1^{0.63} + \gamma_2^* \cdot \sin \alpha_2^* \cdot Z_2^{0.44} \cdot dm_2^{1.63} \cdot l_2^{0.63}}{[(K_1 \cdot \sin \alpha_1^* \cdot dm_1 + K_2 \cdot \sin \alpha_2^* \cdot dm_2) \cdot \mu_{bl}]^{0.37}} \\ &\cdot E^{0.19} \cdot (\eta \cdot \omega_i)^{0.44} \cdot M_{st}^{0.37} \end{aligned}$$

Once $\frac{K'_1 \cdot \sin \alpha_1^* \cdot dm_1 + K'_2 \cdot \sin \alpha_2^* \cdot dm_2}{K_1 \cdot \sin \alpha_1^* \cdot dm_1 + K_2 \cdot \sin \alpha_2^* \cdot dm_2} \approx 1$ and verifying the similarity between the products $\gamma_1^* \cdot \sin \alpha_1^* \cdot Z_1^{0.44} \approx \gamma_2^* \cdot \sin \alpha_2^* \cdot Z_2^{0.44}$ ($2.2 \approx 2.1$) for this case of study, equation (A.33) can be rewritten as shown in equation (A.34).

$$M_{hydrodynamic}^* = (1 - \phi) \cdot \frac{\mu_{EHD}}{\mu_{bl}} \cdot M_{st} + \frac{0.00825 \cdot E^{0.19} \cdot \gamma^* \cdot \sin \alpha^* \cdot Z^{0.44}}{[(K_1 \cdot \sin \alpha_1^* \cdot dm_1 + K_2 \cdot \sin \alpha_2^* \cdot dm_2) \cdot \mu_{bl}]^{0.37}} \quad (A.34)$$

$$\cdot (dm_1^{1.63} \cdot l_1^{0.63} + dm_2^{1.63} \cdot l_2^{0.63}) \cdot (\eta \cdot \omega_i)^{0.44} \cdot M_{st}^{0.37}$$

The torque loss in the TRB, expressed by equation (A.34), shows the importance of the starting torque, generated by the axial pre-load, F_a , applied to the tandem tapered roller bearings. It influences the torque loss on the roller/race contact, which depends on M_{st} and on the product $(\omega \times \eta)$, but it also influences the torque loss on the roller/rib contact, which only depends on M_{st} .

A simplified version of equations (A.29), (A.31) and (A.34) was considered since the starting torque M_{st} already includes significant information concerning the geometry of the tandem TRB's, that is

$$M = M_{transient} + M_{hydrodynamic} \quad (A.35)$$

where

$$M_{transient} = e^{-\frac{M}{\omega_0}} \cdot M_{st} \quad (A.36)$$

$$M_{hydrodynamic} = (M_{st} + C_1)^\alpha \cdot [Cm_1 \cdot (\eta \cdot \omega_i \cdot dm_1^2 \cdot T_1)^{1-\alpha} + Cm_2 \cdot (\eta \cdot \omega_i \cdot dm_2^2 \cdot T_2)^{1-\alpha}] \quad (A.37)$$

The weighting factor ϕ was replaced by a simple version ($e^{-\frac{M}{\omega_0}}$), both depending on the exponential inverse and now the lubricant viscosity and bearing geometry is taken in the main equation. The friction coefficient is function of the lubricant viscosity, rotational speed, bearing geometry and load, parameters that are also considered in the main equation. The constants of equation (A.34) were joined and its variables regrouped, obtaining a simpler equation, which constants and exponents were determined by correlation with the experimental results, section 6.

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Nomenclature

dT_{MER} : roller element elastic rolling torque loss (Nm)
 dT : roller element final torque (Nm)
 dT_{FR} : roller element hydrodynamic rolling torque (Nm)
 dT_{rib} : roller element rib sliding torque loss (Nm)
 Q : hydrodynamic rolling force (N)
 R_o : mean outer raceway radius (m)
 R_i : mean inner raceway radius (m)
 d_m : rolling bearing mean diameter (m)
 M_{rib} : moment due to the rib effect (Nm)
 d : roller mean diameter (m)
 MER : moment due to the elastic rolling effect
 γ : roller bearing geometrical coefficient (–)
 Z : number of rollers (–)
 α' : roller bearing contact angle (degrees)
 E : equivalent Young's modulus (N m²)
 R_s : equivalent radius in the rolling direction (m)
 l : roller length (m)
 U : dimensionless speed parameter (–)
 W : dimensionless load (–)
 η : dynamic viscosity (N s m^{–2})
 ω_i : inner ring angular speed (rad s^{–1})
 F_a : axial load (N)
 F_N : normal load (N)
 μ_{rib} : local friction coefficient (–)
 Q_{rib} : roller-rib load (N)
 f : roller-rib contact height (m)
 b : half of roller-rib contact width (m)
 ν : half of the included roller angle (degrees)
 B : rolling bearing geometrical coefficient (–)
 ϕ_{RIB} : weighting factor for the sliding friction coefficient of the roller-rib contact (–)
 μ_{b1} : boundary conditions friction coefficient (–)
 μ_{EHD} : full film friction coefficient (–)
 μ_e : equivalent friction coefficient (–)
 ϕ_{mer} : weighting factor for the friction coefficient of the elastic rolling effect (–)
 γ' : roller bearing geometrical coefficient (–)
 M_i : single tapered roller bearing torque loss (Nm)
 M_{st} : Roller bearing starting torque (Nm)
 K : roller bearing geometrical coefficient (–)
 K : roller bearing geometrical coefficient (–)
 C : torque loss model constant (–)
 D : torque loss model constant (–)
 θ : global weighting factor for the sliding coefficient (–)
 M : tandem TRB's torque loss (Nm)